Establishment of a methodology for determining stormwater discharges based on hydrodynamic rainfall-runoff sewer network simulations

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Increasingly, pollutant loads emitted from small urban catchments are becoming the focus of centralised and decentralised stormwater management measures due to its significant role in urban water pollution and ecological impacts. Different types of land use, such as roads, residential yards, commercial areas, green spaces, generate different levels and types of runoff with impervious surfaces typically contributing higher pollutants loads. The duration of dry weather phases plays a significant role as pollutant accumulate on surfaces and particulates. The particulate material loads mobilised with the onset of precipitation contribute to a concentrated initial surface runoff (so-called first flush). It carries particulate matter and substances adsorbed (heavy metals, PFAS, pesticides among others). To prevent pollution of aquatic ecosystems the aim of modern rainwater management are increasingly focusing on capturing and treating this stormwater before discharging the rainwater into a receiving watercourse.

In this Master's thesis, the pollutant loads are to be modelled for small urban catchments based on existing measurement results from northern Germany to assess annual pollution loads as mobilized by rainfall events. Similar investigations have been conducted, e.g. by Tuomela et al. 2019 (doi: 10.1016/j.jenvman.2018.12.061) who determined event mean concentrations (EMC) for TSS, TP, TN, various heavy metals for different land cover types using storm water management model (SWMM) of EPA.
The model will be calibrated and evaluated based on quantified concentrations of individual rain events also taking several organic micropollutants into account. The applicability of EMC will therefore also be examined. Alternative modelling approaches with variable concentrations, also taking into account dry weather periods and seasonality, are to be reviewed. Modelling will be conducted using SWMM or Hystem Extran models.
The thesis will therefore primarily comprise a comprehensive data analysis of existing data as a basis for numerical modelling. The results will then be classified as part of a literature study. Treatment concepts will be worked out.

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Firstly, I would sincerely thank my thesis supervisor, Prof Dr. Philipp Zantout-Wilfert and Prof. Dr.-Ing. Kai Wellbrock, for his consistent support and guidance throughout this project. The practical advice and constructive feedback from Professor Kai were crucial in helping me navigate challenges and improve my work. I appreciate the time and effort he invested in reviewing my drafts and offering insights that were both helpful and realistic. I thank him for his steady support and for constantly being approachable when I needed advice.

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Abstract

Urbanization, climate change, and population growth have significantly altered the hydrological and water quality dynamics of urban catchments. This study investigates stormwater runoff quantity and quality in a small commercial urban catchment in Bargteheide, Germany, using the Storm Water Management Model (SWMM). The focus was on Total Suspended Solids (TSS) as a key pollutant, and two rainfall events were selected for model calibration and validation. A detailed sensitivity analysis was conducted to identify key hydrological and water quality parameters affecting model performance. For runoff quantity, impervious percentage and depression storage were found to be the most sensitive parameters. For TSS simulation, maximum buildup and washoff coefficient showed the highest sensitivity. Calibration results showed strong agreement with observed data for both quantity (NSC = 0.84) and quality (NSC = 0.70), while validation results showed satisfactory to poor performance, especially for quality (NSC = 0.30). These discrepancies were attributed to antecedent dry conditions and varying rainfall characteristics. This study demonstrates the applicability of SWMM for stormwater modelling in small urban catchments and highlights the importance of event-specific calibration and high-resolution land use input to improve prediction accuracy.

Keywords: *Rainfall intensity, Runoff quality, Runoff flow, Total suspended solids, SWMM modelling*

Kurzzusammenfassung

Flächenversiegelung, Klimawandel und Bevölkerungswachstum haben die hydrologischen Prozesse sowie die Wasserqualität urbaner Einzugsgebiete tiefgreifend verändert. Die vorliegende Studie untersucht die Menge und Qualität des Niederschlagsabflusses in einem kleinen, kommerziell geprägten urbanen Einzugsgebiet in Bargteheide, Deutschland, unter Anwendung des Storm Water Management Model (SWMM). Der Schwerpunkt lag auf den gesamten Schwebstoffen (TSS) als Hauptschadstoff, für die Modellkalibrierung und validierung wurden zwei Niederschlagsereignisse ausgewählt. Eine detaillierte Sensitivitätsanalyse identifizierte zentrale hydrologische und wasserqualitätsbezogene Modellparameter, die die Modellgüte wesentlich beeinflussen. Hinsichtlich der Abflussmenge erwiesen sich der Versiegelungsgrad sowie die Muldenverluste als die sensitivsten Parameter. Für die AFS-Simulation zeigten sich die maximale Akkumulation von Schadstoffen sowie der Koeffizient zur Beschreibung des Spülstoßes als besonders einflussreich. Die Kalibrierungsergebnisse zeigten eine hohe Übereinstimmung mit den beobachteten Daten – sowohl für die Abflussmenge (NSC = 0,84) als auch für die Wasserqualität (NSC = 0,70). Die Validierung hingegen ergab eine zufriedenstellende bis schwache Modellleistung, insbesondere im Bereich der Wasserqualität (NSC = 0,30). Diese Diskrepanzen wurden auf vorangegangene Trockenperioden sowie variierende Charakteristika der Niederschlagsereignisse zurückgeführt. Die Studie belegt die Anwendbarkeit des SWMM für die Modellierung von Niederschlagsabfluss in kleinräumigen urbanen Einzugsgebieten und unterstreicht die Relevanz einer ereignisspezifischen Kalibrierung sowie der Verwendung hochauflösender Landnutzungsdaten zur Verbesserung der Prognosegenauigkeit.

Schlüsselwörter: *Niederschlagsintensität, Abflussqualität, Abflussmenge, Gesamtmenge an Schwebstoffen, SWMM-Modellierung*

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Abbreviations

SWMM	Storm Water Management Model
RKB A	Study area basin
TSS	Total Suspended Solids
DEM	Digital Elevation Model
NSC/NSE	Nash–Sutcliffe Efficiency Coefficient
R^2	Coefficient of Determination
RSR	RMSE-Observations Standard Deviation Ratio
PBIAS	Percent Bias
EIA	Effective Impervious Area
TIA	Total Impervious Area
EPA/ US EPA	United States Environmental Protection Agency
EU	European Union
NPDES	National Pollutant Discharge Elimination System
QGIS	Quantum Geographic Information System
DIN	Deutsches Institut für Normung (German Institute for Standardization)
EMC	Event Mean Concentration
MB	Maximum buildup
BE	Buildup exponent
WC	Washoff Coefficient
WE	Washoff Exponent
C1, C2	Constants used in buildup/washoff functions
OTT MF Pro	(Device) – Magnetic-inductive flow meter
AFSgesamt	Total Suspended Solids ≥ 0.45 µm (German)
PN1	Rainfall event labels used for simulation (22nd –23rd June 2023)
PN1	Rainfall event labels used for simulation (5th July 2023)

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1. Introduction

Urban stormwater management has become a global concern due to three major trends: population growth, rapid urbanization, and climate change. Climate change alters rainfall patterns worldwide, increasing flood risks in some regions and drought in others (Bashmakov et al., 2022). At the same time, population growth accelerates urbanization, leading to land sealing with asphalt and concrete, which reduces infiltration and increases runoff and flooding. Together, these trends intensify both stormwater quantity and quality challenges (Hong et al., 2019; Gao et al., 2023). Urban runoff does not only give rise to quantity problem but also to quality problem. Rainfall washes pollutants from rooftops, roads, and parking areas into receiving waters. Pollutant loads vary with land cover, surface materials, human activities, and meteorological conditions (Singh et al., 2022). This can degrade downstream water quality, accelerate erosion, and harm aquatic ecosystems. Therefore, effective stormwater strategies must address both water quantity and pollutant quality.

Urban stormwater is generated when rainfall flows over impervious surfaces altered by human activities, reducing groundwater recharge, increasing runoff, and stressing drainage systems, particularly under climate change (House et al., 1993; Barbosa et al., 2012). Effective drainage is essential to protect public health and infrastructure, making it crucial to quantify both runoff volumes and pollutant loads for flood prevention and mitigation planning (Todeschini et al., 2012; Xu et al., 2020). Since large-scale pollutant monitoring is costly and labour-intensive, computational models have become valuable tools to simulate runoff and water quality under different precipitation scenarios (Vezzaro & Mikkelsen, 2012). Model accuracy depends on capturing land use patterns, human activities, and rainfall variability, as these factors strongly influence pollutant generation and transport (Liu et al., 2013). Pollutant-laden stormwater runoff degrades aquatic ecosystems by altering flow regimes, causing erosion, and promoting eutrophication, making water quality management vital for sustainable urban development (Yan et al., 2022).

Water quality models serve multiple purpose, such as the evaluation of potential urban planning strategies for the management of water and pollutant flows in circumstances involving timely decision-making and considerations (Orouji et al., 2013). Usually, model calibration and validation are mandatory before the application of the model for similar conditions and area. The main goal of this modelling is to characterise urban runoff, provide data for analyses of impacts on the receiving water bodies, aid in sizing stormwater infrastructure, analyse quality parameter and finally provide cost benefit evaluation for the water management interventions (Wang and Altunkaynak, 2012).

Though for decades the urban water quality models have been in existence, but their development came into force over the past two decades, mainly due to the implementation of the environmental legislation which required better assessment, prediction and control of urban water pollution. New stormwater quality standards (e.g., European Union (EU) Water Framework Directive (2000/60/CE); United States (US) National Pollutant Discharge

Elimination System (NPDES), Phase I (US EPA, 1990) and Phase II (US EPA, 1999) have highlighted the need to predict the dynamics of pollutant load in urban water quality modelling. These standards have simulated the advancement of many such models, providing theoretical frameworks and challenging applications for the hydrological and environmental engineering research community (Milićević et al., 2010).

The functioning of several stormwater quality models has been reviewed by many, and it is important to calibrate and validate with respect to specific areas to ensure the model correctly represents the real-world conditions (Tsihrintzis and Hamid, 1997; Zoppou 2001). Calibration includes adjustment of model parameters to match the observed data of a specific event, whereas validation helps in testing the performance of the model for separate data. One of the most extensively used urban water quality models is the Storm Water Management Model (SWMM), developed by the Environmental Protection Agency's (EPA). SWMM has been applied and validated across different sections of urban catchments worldwide (e.g., Warwick & Tadepalli, 1991; Temprano et al., 2006; Barco et al., 2008). SWMM model is an active simulation tool that is used to model the stormwater runoff quantity and quality in any urban catchment. It helps in simulating the rainfall interaction with land surfaces, its flow through drainage networks and the transportation of different pollutants in a single framework. It is an open source, its license is free of charge, it is flexible for working in different land use and various storm event, and its extensive documentation makes it a preferred tool over other commercial software for both practical as well as research applications.

Though thorough field data collection is often expensive and time consuming, models allow for long-term simulations using lesser data and provide simulations beyond the time intervals of monitoring programs (Vezzaro & Mikkelsen, 2012). A variety of models has been established to address both quantity and quality of the stormwater (Niemczynowicz, 1999). Despite this progress, knowledge gaps remain. Quality models rely on simplified pollutant build-up and wash-off functions, assuming pollutants accumulate during dry periods and are mobilized during rainfall (Dotto et al., 2010). These functions work reasonably well at small scales but become uncertain in large heterogeneous catchments, where parameters are often averaged and may oversimplify surface behaviour. This limits the reliability of stormwater quality predictions and highlights the need for better parameterization across different environments (Hossain et al., 2010).

This thesis addresses these gaps by focusing on stormwater quality parameters. It investigates pollutant build-up and wash-off dynamics in urban catchments and derives parameters that improve stormwater quality modelling. By extending knowledge beyond small-scale assumptions, this work contributes to more reliable modelling and planning for large urban catchments.

1.1 Objectives of the study

The prime aim of this study is to simulate pollution loads which occur during storm events. For this, calibration of a SWMM model for the water quantity and quality for a stormwater in a small German urban catchment is necessary. The more thoughtful idea to do this study was to gain more knowledge about both the hydrograph and pollutograph model and how well it can function for several events. The overall idea is to use a calibrated model to predict/ estimate annual pollutant loads.

The following detailed objectives were necessary to get the final objective:

1. Set-up hydro-numerical model of small urban catchment(s) and calibrate for singular storm events.
2. A sensitivity analysis would be conducted along with the calibration and validation of the model.
3. Check model for applicability to predict pollutant loads only for TSS (for single storm events).
4. If satisfactory results for TSS loads had been gained, also model other types of pollutants which are accumulation during dry-weather periods.

These objectives should be achieved by using data from a monitoring program which aimed to quantify pollution loads in surface runoff during storm events.

2. Literature review

2.1 Urban hydrology

Stormwater runoff occurs when the rainfall water does not infiltrate or percolates within the ground due to reduction in pervious surface over the years (US EPA, 2014b). The stormwater is specifically called urban stormwater runoff due to its flow in the urban areas if these flows in other areas they might be referred by different names. The increase in urbanization has drastically changed the water flows way, rate of evaporation and infiltration impacting the original hydrology of the catchment. This impact varies depending on the percentage of impervious surface in different areas (Figure 1a). Hence, the stormwater runoff is lesser in rural or forest areas when compared to the urban areas that comprises of lots of impervious surfaces such as roads, parking lots, concrete roofs, sidewalks, playfields, etc. Though not all the area are converted into impervious surfaces but still the areas that are of natural surfaces are very compacted which again reduces the total infiltration capacity of the area.

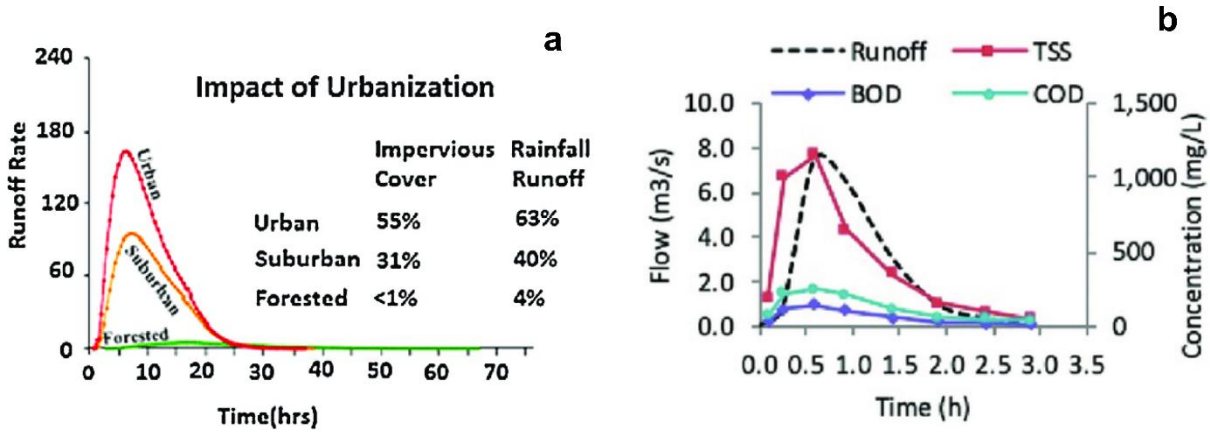

Figure 1. *a) Hydrographs showing runoff from 3 test areas at different levels of development (based on impervious cover). The hydrograph illustrates the impact of urbanization on runoff (Blair et al., 2011) b) Hydrograph and Pollutograph for high rainfall intensity (Kumar et al., 2020).*

As a result of reduced infiltration, the rainfall runoff speeds up leading to surge of direct and total runoff volume, increased peak flows, and rapid concentration time and recession time (Fletcher et al., 2013; Barbosa et al., 2012). When compared the effective transport of runoff water, most impervious surfaces (artificial surfaces) do it more effectively than the natural surfaces. The reason behind this is the hindrance due to the natural surfaces which mostly is absent in the impervious surfaces. For instance, if there is a grass field in one area and the asphalt road in another area, the runoff water will move slower in the grass field due to obstacles which is absent on the absent surface making it smoother for the runoff. The smooth surface of the impervious surface is the reason why the stormwater rapidly flows to the low-lying areas, pits, inlets and the stormwater infrastructure inhibiting the infiltration (Shuster et al., 2005). The rapid collection of stormwaters sometimes results in urban flooding at the catchment's outlet or somewhere in the pipe network if the capacity exceeds or if the runoff

has insufficient time to discharge. Due to rapid water movement and flooding, the downstream areas of the catchment might also suffer terrible stream bank erosion issues (Sillanpää, 2013).

The effective impervious area (EIA) refers to the total portion of the impervious area that is hydraulically connected to the drainage system and contributes mostly during the most rainfall event's surface runoff (Shuster et al. 2005). Urban areas comprising larger EIA is concerning because even if the size of rainfall events are small or medium they might cause significant harm due to their frequent occurrence. When Booth et al., 2002 compared the 10-year relapse interval in an area before urbanization and 2-year relapse interval in the same area after urbanization with 10% level of EIA he found out that the runoff generation of both the areas capitulated the same amount of discharge. When measuring the larger rainfall events, it is important to consider the total impervious area (TIA) because the separate impervious area also contributes to the total flow (Sillanpää, 2013).

The rise of urbanization has caused reduction in groundwater recharge by hindering infiltration, moreover it is also capable of impacting the groundwater quality if the quality of stormwater is degraded (Shuster et al., 2005). The transformation of pervious surfaces into impervious surfaces, urban microclimate and decline in infiltration all together can impact the evapotranspiration too. Though, the evapotranspiration study in the urban areas have not been so much explored but recently started to gain more light with the inspiration to know more about how urban water balance are impacted by the stormwater management (Fletcher et al. 2013). Different characteristics of the catchment display different effect on the runoff flow. The increase in impervious percentage of the area is significant it increases the runoff volume as well as the runoff peak. The width of the area effects the shape of the flow graph it decreases the runoff volume and increases the runoff peak. For rainfall event of variable intensity, increase in the width tends to produce earlier and higher hydrograph peaks, in general faster response (Figure 1). The slope of the area also effects the shape of the flow graph it decreases the runoff volume and increases the runoff peak. Roughness displays opposite effect as the width on the shape. Depression storage has moderate effect on the runoff, shows significant effect for low depth rainfall events (Rossman, 2010).

2.2 Urban runoff water quality

2.2.1 Pollutants and their impact

Pollutants are generated by the anthropogenic activities on the surface of urbanized areas which is carried into the receiving streams via the stormwater runoff impacting the water quality of natural water body. The major pollutants impacting the quality of stormwater are suspended solids, bacteria, heavy metals, and nutrients. Pollutants enter in the stormwater through different sources. Main sources are the commercial and residential zones, roads, traffic, parking areas, driveways, sidewalks, pesticides, atmospheric deposition, industrial activity, and many more artificial structures. Nutrients in the pollutants can be varying but mostly it is phosphorus and nitrogen in various forms. Excess of these nutrients may lead to eutrophication in downstream water by increasing macrophytes and phytoplankton growth. These issues can lead to change the ecosystem structure and disrupt the species diversity (Burian et al., 2001).

Another major pollutant present in the stormwater are the heavy metals (Barbosa et al. 2012). Commonly occurring heavy metals are lead, zinc, copper and nickel (Sillanpää, 2013),

where traffic being the main contributor of lead and zinc (Table 1). Other source of heavy metals could be industrial actions and spill, atmospheric deposition and corrosion of building (Herngren et al. 2005). Heavy metals are transported into downstream via stormwater runoff by attaching to the finer particles of the suspended solids. Besides, the above-mentioned pollutants the stormwater runoff also include bacteria and microbiological pollutants which originates from the degradation of organic waste such as animal faeces, soils or vegetation. When these harmful bacteria, virus or parasites enters the downstream water it can be harmful to the living entity (human or animals) causing illness (James and Joyce, 2004). Hence, it is prohibited for humans to use the water enriched in microbial pathogens for recreational activities (Mallin et al., 2009).

These pollutants impact the ecology of the water in various ways whose effect can be cumulative, acute or chronic. Though acute effect is for short term it can cause serious damage, for example spilling of large chemical or oil in water have acute impact but it can cause major fish kills. Continuous build of these pollutants results in cumulative contamination in the downstream water. The impact of these pollutants is also size dependent along with being time dependent. If the receiving water body (pond) is small, it will show faster response when compared to a large water body (lake or sea). Barbosa et al. 2012 mentioned how different pollutants have different forms in the stormwater such as heavy metals are present dissolved form, whereas organic matter are present in colloidal form many other can be found attached to the suspended particles. Hence it is important to mention all the forms when planning the management practice.

Table 1. *Sources of some contaminants in urban storm water runoff (US EPA, 1999)*

Contaminants	**Sources**
Sediments	Roads, driveways, streets, lawns, atmospheric deposition, construction activities, etc.
Nitrogen and Phosphorus	Animal waste, detergents, soil erosion, automobile exhaust, atmospheric deposition, lawn fertilizers.
Oil and grease	Vehicle maintenance areas, illicit dumping to storm drains, parking lots, driveways, roads, gas stations.
Pesticides and herbicides	Residential lawns and gardens, soil wash-off, industrial and commercial landscaped areas, roadsides.
Metals	Corroding metal surfaces, soil erosion, industrial areas, combustion processes, atmospheric deposition, bridges, automobiles.
Organic materials	Animal wastes, commercial landscaping, residential lawns and gardens.

The discharge of the runoff into downstream water body does not only have impact on the quality but also have impact on the geomorphology (House et al., 1993). If the peak of runoff is high, it could cause channel migration or erosion of the downstream bank. These eroded streambanks can also have impact on the temperature regime of the downstream river, which may also impact the ecosystem. The rapid growth of urbanization has reduced the coarse

sediments increasing the smaller particles flowing and depositing in the downstream, resulting in reduction of diversity in the water body (Barbosa et al. 2012, Fletcher et al. 2013).

2.2.2 Total Suspended Solids

Most of the pollutants including the nutrients, heavy metals and pesticides are present in dissolved form attached to the suspended solids, which makes the total suspended solids (TSS) concentrations the fundamental indicator pollutant in this modelling (Akan & Houghtalen, 2003; Egodawatta et al., 2007). TSS may signify other pollutants only if their physico-chemical traits are similar such as heavy metals, phosphorus and hydrocarbons, but it cannot represent dissolved pollutants like nitrogen and other soluble substances because their physico-chemical traits are different (Miguntanna et al., 2013). Since, the TSS is a significant pollutant in the downstream river, in this study we will use TSS concentration as a measurement of the water quality.

TSS concentration have many impacts on the water body apart from the increase in contaminants, such as its increase in concentration can even alter the physical and chemical properties of the water bodies. High concentration of TSS can lead to dissolved oxygen depletion, reduction of light penetration which can reduce the photosynthesis in aquatic plants, temperature changes which can form layers in water bodies, and might fill the reservoirs and channels reducing the water quantity (Bilotta & Brazier, 2008). These changes can cause increased costs of water treatment, lead to reduce the lifespan of hydraulic structures and degrade the water quality.

Table 2. *Typical pollutant loadings from runoff by urban land use (Rossman, 2010)*

Land use	TSS (lbs/acre-yr)
Construction	6000
Park	3
Industrial area	860
Parking	400
Commercial area	1000
Low density area	10
Medium density area	190
Low density area	420

Preferably the TSS loads of any catchment are calculated by considering the relationship between the TSS concentration and runoff volume, both being constantly recorded. But it is not feasible to constantly record the TSS concentration in the water, as it is very tedious and time consuming (Linjama et al., 2009). Alternate options such as are also available but they are expensive and untrustworthy (Grayson et al., 1996).

2.2.3 Characteristics of surface buildup and wash off

The buildup, wash off and the generation of the pollutants in any catchment are highly influenced by many factors such as the impervious area percentage, climatic conditions and geography of the area (Liu et al. 2012). To understand the surface buildup and wash off it is

important know the hydraulic connection of the drainage system and how the urban features are distributed spatially which defines the how the rainfall responses to the catchment (Liu et al., 2012; Sillanpää, 2013). This response of the catchment also impacts the water quality of the downstream. Most of the area in urban surface are wrapped by the transport related works, almost 63 to 70% at commercial and residential areas. These road surface plays an important role in causing the quality issue in the stormwater (Egodawatta et al., 2009). Due to regular operation of vehicles on these roads the surface may undergo abrasion or sometimes leakages of oils from the automobiles can happen then all these are washed off by the rainwater. Mostly the roads are sloped so that all these stormwaters can drain into downstream making the flow of stormwater faster and release of pollutants easier.

Compared to other impervious surfaces roads have higher accumulation of pollutants as well as pollutant buildup rate (Egodawatta & Goonetilleke, 2008a). But the pollutants from the surface of roads cannot be washed off easily as their surface have coarser structures whereas roof or other structures have comparatively finer surface (Egodawatta et al., 2009). Since the pollutants from the roofs can be washed off even during little rain, roofs become a significant source of the water quality problems in the downstream. The roof area estimated by Egodawatta and Goonetilleke (2008a), covers almost 19, 23 and 38% in three urban area, which they distinguished as residential area. These origins of these pollutants are generally the atmospheric deposition (Egodawatta et al. 2009) or the wearing of the roof materials (Van Metre & Mahler 2003) due to the acidity of rainfall, which is why they are comparatively finer in size (Egodawatta et al. 2009).

The characteristic of buildup depends on the immediate land use activities, climate conditions and traffic which determine the quantity and quality of the pollutants in that area. After a high rainfall event (storm event) the buildup rate is higher but gradually reduces as the dry period starts. That is why during the first seven dry days after the storm event 80% of the total pollutant buildup (Egodawatta et al., 2009). Wash off varies depending on the density of the particle (Egodawatta et al., 2009) and the size of the particle (Miguntanna et al. 2013) and these in turn are dependent on the characteristics of the impervious surface. Hence, the wash off of pollutants is more dependent on the structure and type of the catchment (impervious percentage, slope, etc.) instead of the pollutant available initially (Egodawatta and Goonetilleke 2008b).

2.3 Stormwater quality modelling

With rapid urbanization, it is increasingly important to study stormwater quality and develop computational tools to predict pollutant dynamics. Beyond flood prevention through stormwater diversion, the potential for stormwater reuse for example rainwater harvesting makes water quality assessment even more relevant. For example, roofs are a major source of pollutants in stormwater, and as they are often used for harvesting, their runoff quality is directly linked to reuse applications (Egodawatta et al., 2009). More broadly, the quality of urban runoff determines its suitability for reuse, treatment needs, and its environmental impact

when discharged. Climate change must also be considered, as stormwater quality and pollutant mobilisation are highly dependent on rainfall intensity and patterns (Fletcher et al., 2013).

The stormwater quality modelling is a mathematical technique which portrays the change that the quality of stormwater undergoes because of rainfall and runoff effect in any catchment area (Egodawatta & Goonetilleke, 2008b). In this modelling everything including buildup, wash off and transport processes of pollutants are presented by mathematical equation systems. Modelling of stormwater quality is a complicated process as it comprises of multiple factors and hence demands for simplification during its construction. Pollutant buildup and wash off being the main factor used in quality modelling vary extensively in varying environments for different impervious surfaces. Nonetheless, the data is constrained, and all things cannot be considered therefore we require generalized approximate values for modelling (Obropta & Kardos, 2007).

Land use that is used widely to present pollutants in urban stormwater generated on the surface are grouped in a single class, that is residential, commercial or industrial areas. Even inside the similar land use pollutant processes varies because the area-specific properties perform crucial role in formulation of the pollutant buildup and wash off (Liu et al., 2012). Since the pollutants generated in the storm water from different impervious surfaces has not been broadly studied, the accretions and generalization of these types are unavoidable (Brodie & Dunn, 2010). This is why this modelling is constrained to common level and considers grouped land uses. Till date no work has been done utilizing high spatial resolution for modelling of stormwater quality. As this approach requires significantly more data and computational effort, it has the potential to better capture spatial heterogeneity in land use, surface materials, and pollutant sources. These factors strongly influence pollutant build-up and wash-off processes, meaning that higher resolution may lead to more accurate and transferable parameterisation compared to models using averaged inputs.

The calibration for large catchment model for small definite area (Egodawatta & Goonetilleke, 2008a) or for the area with multiple land uses (Temprano et al., 2006) has been worked on. However, studies on large catchments with multiple land uses have so far been limited. In this we have considered a small area of commercial land to study which can in future be used as a reference for modelling purposes of similar areas. Several studies have been done between the relationship of total suspended solids and other event mean concentration with various types of rainfall runoff properties for given type of surfaces e.g., roofs, highways and roads (Gilbert & Clausen 2006). But operating EMC only have some limitations especially for the first flush phenomenon. Therefore, for this study we will be using the exponential function to understand the fluctuations dynamics in water quality relating the concentration of pollutant nonlinearly to runoff.

2.4 Drawbacks of the measurements

Uncertainty in modelling is very common, and it can be due to many reasons such as error during modelling or during observations, sometimes even due to some phenomenal changes. Since observations only represents reality, it might have some level of uncertainty

(Gandin 1988). Estimation of these uncertainties and understanding the probable errors is an important task to make the urban stormwater management safe and reliable. The stormwater quality measurement equipment is prone to contamination, gases, changes in pressure and temperature, ice, etc., causing break down during monitoring (Linjama et al., 2009). There are two types of data errors i.e., systematics and random errors. Systematic errors occur due to the equipment malfunction or errors in data procession, programming or reception, whereas the errors produced by assembly of independent factor and are clear are random errors (Gandin 1988).

In stormwater studies, a major source of uncertainty originates from measurement errors, which then propagate into model calibration and validation. Runoff, rainfall, and water quality are often measured at the catchment outlet, which is treated as representative for the entire catchment. This simplification introduces errors, as rainfall measured at a single point may not reflect the true spatial variability of precipitation across the catchment (Niemczynowicz, 1999). Additional errors may occur with tipping-bucket rain gauges, due to splashing, wetting, wind effects, or equipment malfunction. Runoff measurements can also be biased if sewer flows are influenced by anthropogenic activities unrelated to precipitation. Similarly, TSS monitoring is prone to systematic errors. For example, blockages caused by litter or sediment deposition can lead to artificially high readings, while dirt accumulation on sensors may cause drift. Although modern TSS sensors include automatic cleaning systems, data verification and inspection remain essential to ensure accuracy (Fletcher and Deletic, 2008).

3. Methodology

3.1 Study area

The study area is in Bargteheide in northern part of Germany. This area is characterized commonly by commercial areas with size approximately of 21.087 ha. The percentage of impervious surfaces on an average is 61.88% whereas the mean slope is 0.82% (Figure 2 and 3). The average yearly rainfall is 64.1 mm with highest amount of rainfall in July, August and December. The elevation range in the study area is almost flat, ranging in value from 36.73 m in the north-east to 41.78 m in the south-west.

Figure 2. *Aerial photograph of the study area (Wellbrock et al., 2024).*

Figure 3. *Open street map (Pink colour: paved areas, grey colour: buildings and green colour: pervious areas,* ***but in real it can vary****) with the nodes and the conduits dividing the catchment into subcatchments which will be used for the modelling purpose (Raw images from Wellbrock et al., 2024).*

3.2 Storm event and Runoff quantity and quality data

The data used in this study were collected by Wellbrock et al., 2024 for two different events i.e., first on 22nd and 23rd of June 2023 and second on 5th of July at the inflow of storm water treatment basin. The precipitation was collected using a stainless-steel sheet (1000x2000mm) which was bent concavely to collect the precipitation which was further collected into a stainless-steel collection vessel. Both vessels were cleaned with acetone before collection and were dried with compressed air to avoid sample contamination (for more details refer to section 3.3, Wellbrock et al., 2024). The flow was measured at two places, but we will be using only inlet flow data for our work. Flow at inlet was measured using a device called OTT MF Pro, which is a magnetic-inductive flow meter. This device helps in measuring the velocity of the water at different points in the pipe (for more details refer to section 3.4, Wellbrock et al., 2024). The rainfall intensity was measured before both the event to know about the antecedent dry days (Figure 4 and 5). The variation of flow runoff during both the events with respect to the precipitation has also been shown in Figure 6 and 7.

In our present study to understand the quality of the stormwater runoff we will be analysing Total suspended solids (TSS) as pollutants. The AFS_{gesamt} (total suspended solids ≥ 0.45 µm) was determined following DIN 38409-2 by filtering homogenized stormwater samples through a 0.45 µm polyether sulfone (PES) membrane without prior sieving. The sample was first mixed using an Ultra-Turrax at 10,000 rpm for 2 minutes, then vacuum filtered. The filter with retained solids was dried at 105 °C for one hour, cooled in a desiccator, and weighed to determine the mass of suspended solids. The concentration (mg/L) was calculated based on the mass difference and sample volume (typically 250–1000 mL). This method ensures all particles larger than 0.45 µm are measured, with a quantification limit of 2 mg per filter (for further details refer to the section 3.7, Wellbrock et al., 2024).

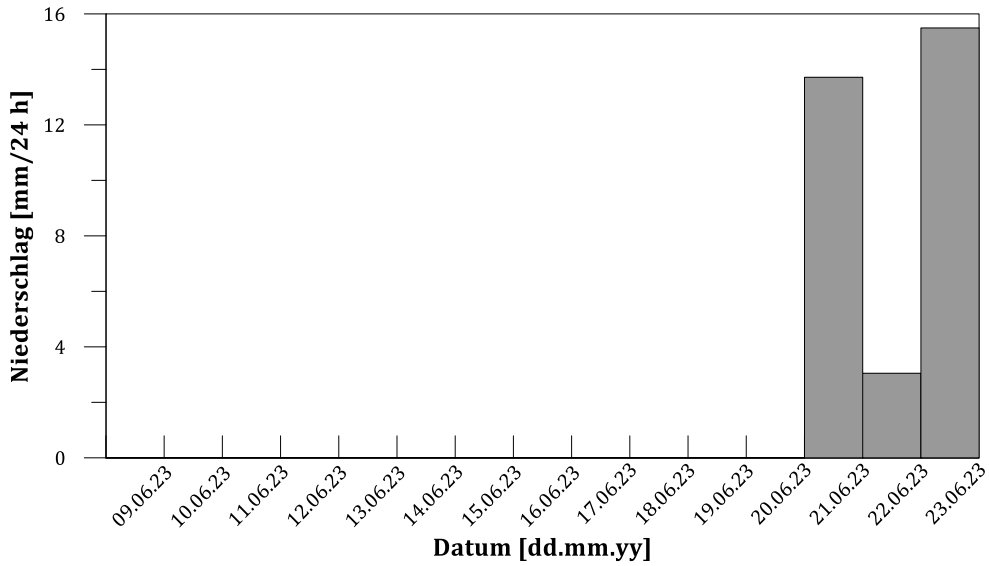

Figure 4. *24-hour precipitation data from a weather station close to RKB A 14 days before the precipitation event from June 22, 2023, to June 23, 2023.*

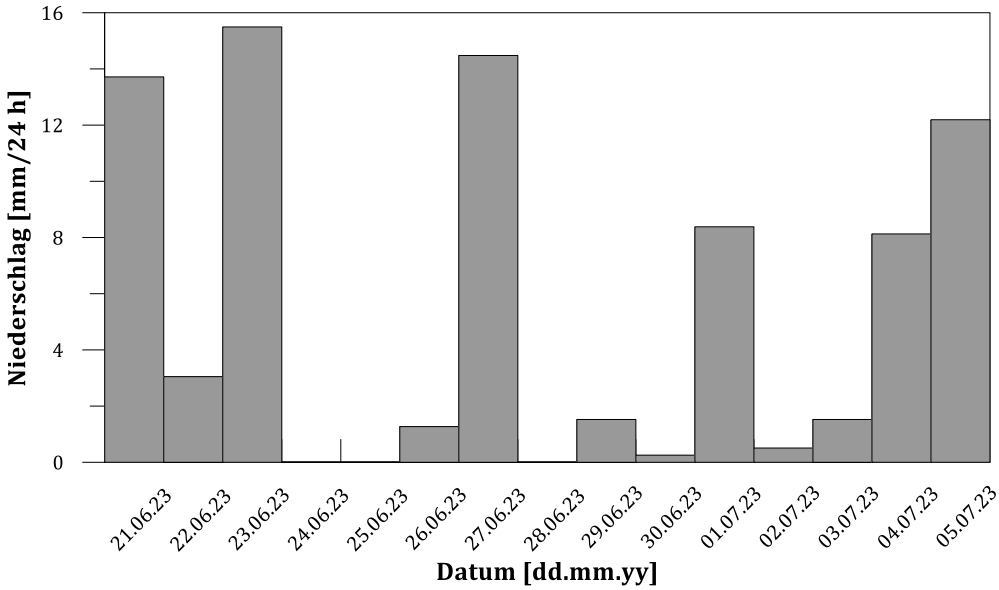

Figure 5. *24-hour precipitation data from a weather station close to RKB A 14 days before the precipitation event on July 5, 2023 (adapted from Wellbrock et al., 2024).*

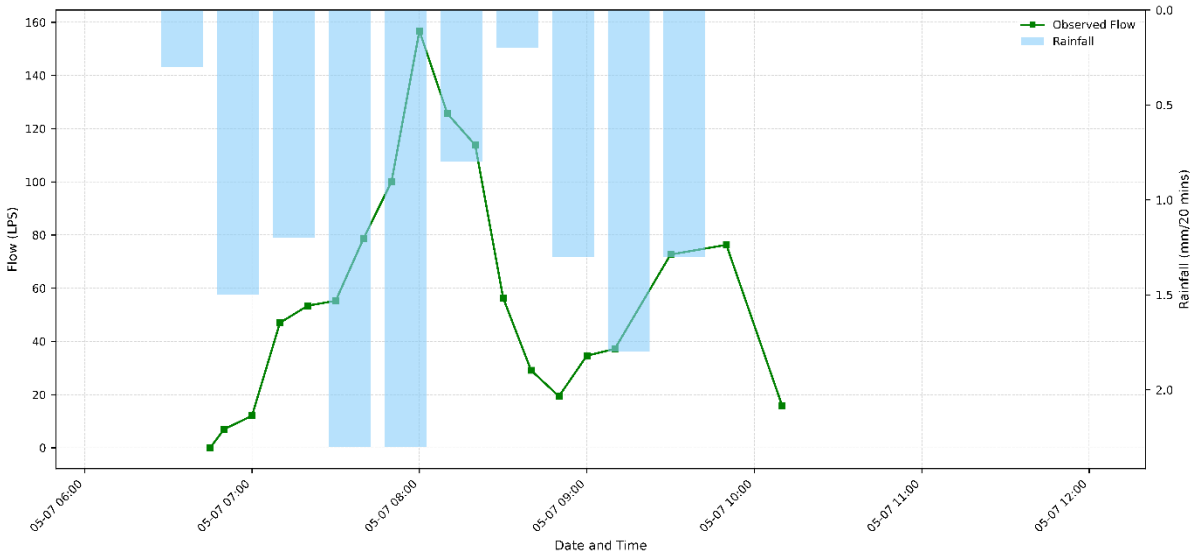

Figure 6. *Graph of flow hydrograph (line) and rainfall bar graph for PN2 rainfall event (5th July 2023, adapted from Wellbrock et al., 2024)*

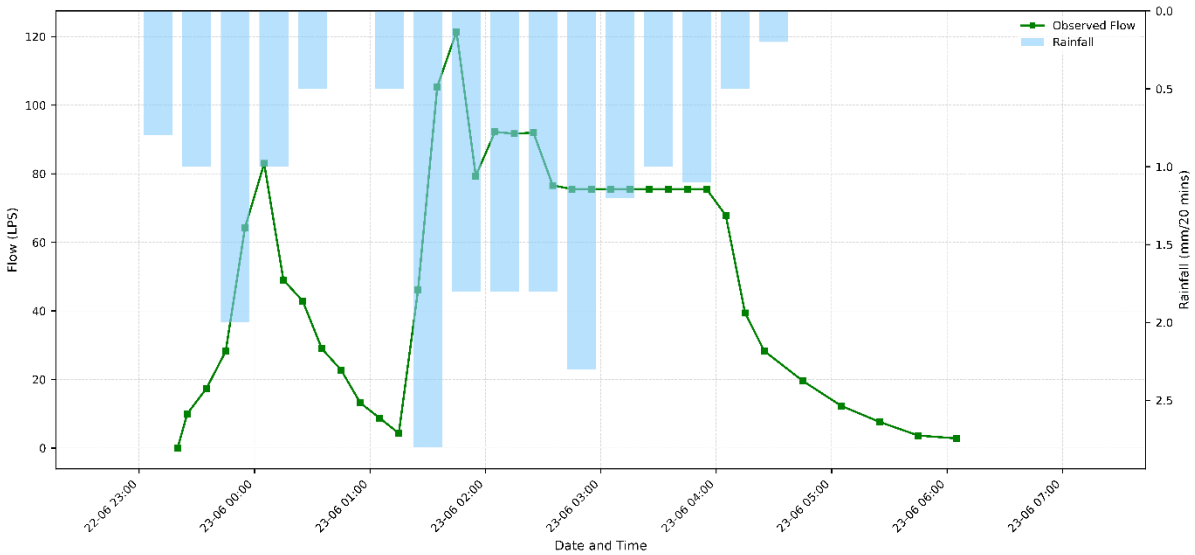

Figure 7. *Graph of flow hydrograph (line) and rainfall bar graph for PN1 rainfall event (22nd and 23rd June 2023, adapted from Wellbrock et al., 2024)*

3.3 Model Parameterization

For this study the stormwater management modelling software (SWMM) was used. This model helps in developing dynamic model for rainfall-runoff simulation which can be used to simulation runoff quality and quantity of single rainfall events or continuous events in any urban areas (Rossman, 2010). The input parameter used in this study are combination of model default values, literature review, field data and existing datasets. For this study the catchment was divided into 25 subcatchments to have a detailed quantization of the manhole's distribution and catchment. The separation of subcatchments is based on land use and drainage network. The general process model used in the simulation were the rainfall/runoff, flow routing and water quality. The routing model used for our study is kinematic wave, the infiltration model selected is modified Green-Ampt. Since the soil texture was sandy loam of

this area the suction value was 4.33 inches and saturation hydraulic conductivity was 0.43 inches. Evaporation was given constant value of 3mm/day. Other values for options were kept as default values.

The physical parameters of the subcatchments such as elevation, area, width, impervious percentage and slope were measured from the digital elevation model (DEM) in QGIS software. The parameter for conduits and nodes such as invert elevation, max depth, length, shape and material of the pipes, were extracted from the sewage network map provided by the municipal office. Afterwards, all the measured or extracted values were imported in the software to start the simulation, followed by connecting nodes, conduits and subcatchments to start the simulation. Values for infiltration were calculated from the manual assuming the soil to be sandy loam in nature.

The width of the subcatchments is calculated from the aerial photographs in QGIS map and it is the real width of the overland flow running down the slope in an idealized, rectangular catchment. Since the real subcatchment are not rectangular we should use other methods to measure the width of the subcatchments (Rossman, 2010). The change in the width affects the hydrograph in such a way that increase of the width tends to originate peak flows to occur quicker.

The slope of the subcatchments is also calculated using the aerial photograph it is simply the calculation of difference of elevation divided by the flow length (Rossman, 2010). Manning's roughness coefficients for the overland flow have been taken from Engman (1986). Depression storage is a storage that essentially should be filled preceding the incidence of runoff on both impervious and pervious areas (Viessman and Lewis, 2003). It is derived from the rainfall-runoff data for impervious areas using the formula:

$$V = C(P - d_s) \tag{1}$$

Where V is the runoff volume, P is the rainfall volume, d_s is the depth of depression storage and C is the coefficient. For the infiltration the Green-Ampt method was used, soil texture classified as sandy loam (Rawls et al., 1983).

3.4 Water quality

The hydrological modelling and water quality modelling in the SWMM modelling is not dependent on each other hence the calibration and validation should be done distinctly for both the models. The simulation of buildup and wash off of the pollutants is used for modelling the quality of stormwater in any catchment area. The buildup function is the measure of the amount of pollutants accumulated on any impervious surface during the dry season meanwhile the wash off function measures the amount of pollutants that are washed off after the rain event. There are three different types of pollutant buildup function available in SWMM software (Power, Exponential and Saturation) and the user can use any of the above.

Power function buildup designates how the mass of the pollutant buildup (B) accumulates, until a maximum limit is achieved, presented by the formula:

$$B = Min(C_1, C_2 t^{C_3}) \tag{2}$$

where C_1 represents max build-up (mass per unit area or curb length), C_2 represents rate constant of build-up function, t presents time elapsed and C_3 is a time exponent. The time exponent (C_3) is directly proportional to the buildup of pollutants. Similarly, the value of rate constant $C2$ is directly proportional to the time taken to reach the maximum. C_3 should always be less than 1 so that with increase of time the buildup rate decreases. Hence when the value of C_3 is 1 the buildup function varies linearly.

Exponential buildup approaches a maximum limit asymptotically following exponential growth curve and represented by the following formula.

$$B = C_1(1 - e^{-C_2 t}) \tag{3}$$

where C_1 represents max build-up (mass per unit area or curb length), C_2 represents rate constant of build-up function (1/d), and t represents the elapsed time (d).

The saturation function describes a build-up that begins linearly and further proceeds to declines until a saturation value is reached, it is represented by the following formula.

$$B = \frac{C_1 t}{S+t} \tag{4}$$

Figure 8 presents and compares the graph of different buildup functions using hypothetical pollutant (Rossman, 2010). For our study we used Power buildup function.

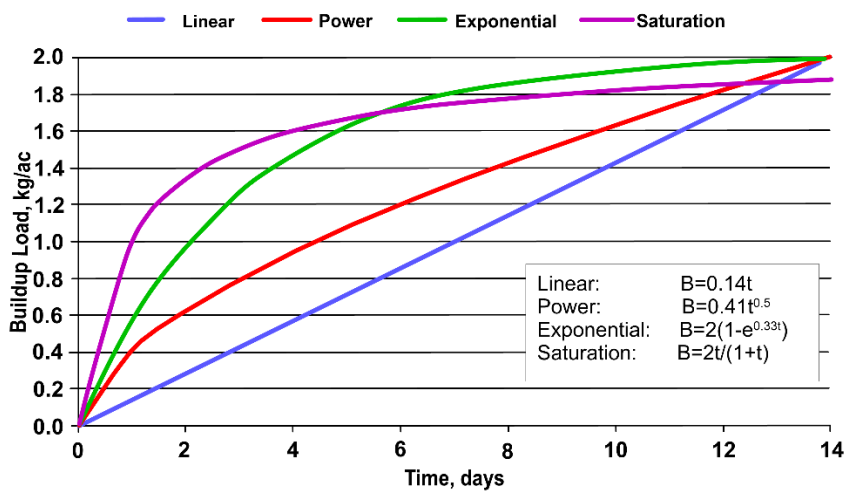

Figure 8. *The buildup curves of different functions for a hypothetical pollutant*

The process involving erosion and dissolution of components from any roof, road or other surface of the subcatchments during the rainfall-runoff event is called Wash off. The

properties related to the wash off of pollutants are usually defined by an exponential function, rating curve or event mean concentration (EMC).

The exponential wash-off function develops the exponential relationship with wash-off load described by the formula:

$$W = C_4 q^{C_5} B \tag{5}$$

where C_4 represents wash-off coefficient (WC), q is the runoff rate per unit area (mm/hr), C_5 is the wash-off exponent (WE), and B is the total mass of built-up pollutant which is measured in mass units unlike the buildup functions.

Rating curve wash-off depicts the proportional relationship between wash off load and the runoff raised to a power, represented by the following formula:

$$W = C_4 Q^{C_5} \tag{6}$$

where C_4 represents WC, Q is the runoff rate in flow units and C_5 is the WE.

The EMC wash-off uses the same formula as the rating curve function where WE value is 1, and WC concentration in kg/l while choosing SI units. Figure 9 shows the curves of the different washoff function and the compares the shape of runoff pollutgraphs having initial buildup of 20lbs pollutant in an acre of catchment subjected to a 2-inch, 6-hour storm with a triangular-shaped runoff hydrograph (Rossman, 2010).

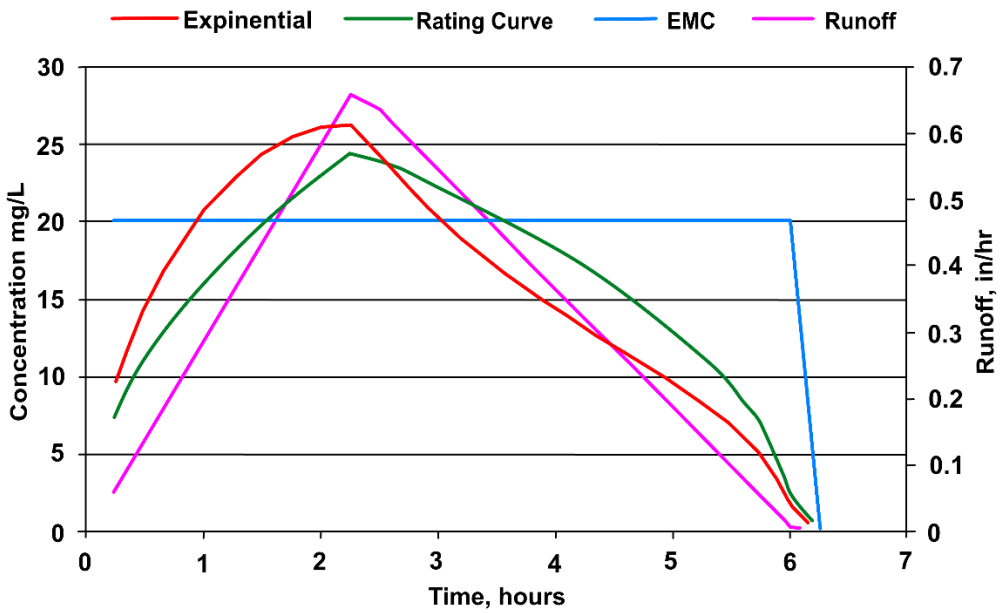

Figure 9. *Different washoff functions comparison curve*

For our study we will be using exponential function as it can produce a nonlinear wash off curve and is dependent on the accumulated buildup. The nonlinear wash off curve is produced by rating curve too but it is not dependent on the buildup. The only function which

produces linear curve and is also independent from the buildup is the EMC wash off function, in other words it displays only simulated flow multiplied by the predefined concentration.

The quantity of pollutants can be either given to the model as input directly at the starting of the simulation or can be calculated according to the dry days number defined by the user before starting the simulation. In case there were no dry days then the buildup will start the accumulation from zero at the beginning of analysis. For our study we set the number of dry days as zero as there was rainfall event every day.

The user can mention different land use category such as commercial, residential or industrial in the model manually. When the resolution is even coarser user can add more detailed land use such as roof, road, pavements, parking lot etc. In this study we have used smaller resolutions hence divided the subcatchments into only roads and commercial areas. Different land uses can be given different or same buildup and wash off functions but the parameters for the functions should be different for different land use type depending on their contribution to the pollutant amount. In case you apply same function as well same parameters then the pollutant buildup and wash off in each catchment will be similar. For our study we have given different buildup (Power function) and wash off (exponential function) function for the two land uses (both impervious). For pervious surface we did not give any buildup or wash off function as they do not contribute to the pollutant loads.

There are other options available that the user can define in the model to get more precise results is the pollutant removal efficiencies linked to the best management practices and street sweeping data for different land uses. Since we had no data regarding these, we neglected this information for our study.

For the study of water quality, the pollutant that needs to be consider in this study are provided a name, the concentration in which it needs to be measured in the model. Other information we which did not add but can play important role are pollutant concentrations in rainwater, groundwater, infiltration/inflow or in dry weather sanitary flow. Which means the pollutant not only comes from the buildup and wash off but can also originate from rainwater, groundwater, etc. The concentration of pollutant buildup and wash off varies with land use but the pollutant concentration in rainwater, groundwater, etc remains same throughout the catchment. We can also use some pollutants as Co-pollutant to other if their concentration is a fixed fraction of another pollutant concentration. Since in this study we only defined TSS as the pollutant type, we did not use the Co-pollutant term. Also, we only used the buildup and wash off on the surface therefore all other factors were given a value of zero.

3.5 Goodness of fit test

The model performance through the calibration and validation process were firstly estimated graphically and then statistically using goodness of fit indices (Chow et al., 2012; Krebs et al., 2013; Moriasi et al., 2007). Four statistical measures considered for evaluating the model performance include the Nash–Sutcliffe efficiency coefficient (NSC), coefficient of

determination (R^2), RMSE-observations standard deviation ratio (RSR) and percent bias (PBIAS). The formulas for these are given below:

$$NSC = 1 - \frac{\sum_{i=1}^{n}(Obs_i - Sim_i)^2}{\sum_{i=1}^{n}(Obs_i - \overline{Obs_i})^2} \tag{7}$$

$$R^2 = \left[\frac{\sum_{i=1}^{n}(Obs_i - \overline{Obs_i})^2(Sim_i - \overline{Sim_i})^2}{\sqrt{\sum_{i=1}^{n}(Obs_i - \overline{Obs_i})^2 \sum_{i=1}^{n}(Sim_i - \overline{Sim_i})^2}}\right]^2 \tag{8}$$

$$RSR = \frac{\sqrt{\sum_{i=1}^{n}(Obs_i - Sim_i)^2}}{\sqrt{\sum_{i=1}^{n}(Obs_i - \overline{Obs_i})^2}} \tag{9}$$

$$PBIAS = \frac{\sum_{i=1}^{n}(Obs_i - Sim_i) \times 100}{\sum_{i=1}^{n} Obs_i}\% \tag{10}$$

where Obs_i is the ith measured values for the component being estimated; $\overline{Obs_i}$ is the mean of measured values for the component being estimated; Sim_i is the ith computed values for the component being estimated; $\overline{Sim_i}$ is the mean of computed values for the component being estimated; n is the total number of values measured. Moriasi et al. 2007 conducted the interpretation of the goodness-of-fit parameters for the calibration and validation stages of the model. Table 3 presents the qualitative performance rating system employed to evaluate the performance of the SWMM model.

Table 3. *Model performance rating system (Moriasi et al., 2007).*

Statistical Criterion	Value Range	Model Performance Rating
NSC	>0.65 >0.50 ≤0.50	Good Satisfactory Unsatisfactory
R^2	≥0.8 ≥0.6 <0.6	Good Satisfactory Unsatisfactory
RSR	≤0.55 ≤0.70 >0.70	Good Satisfactory Unsatisfactory
PBIAS	≤ ±15% ≤ ±25% > ±25%	Good Satisfactory Unsatisfactory

3.6 Sensitivity analysis

To understand how sensitive our model is for the parameter of interest we conduct sensitivity analysis. It is often done before the calibration to get in depth knowledge of the structural dependability of the model and is helpful to decode the suitable parameters that can be used for the calibration (Mustonen 1986). In case the model is least sensitive for specific

parameters, they should not be included in the calibration. Meanwhile highly sensitive parameters should be included in the calibration.

Sometimes researcher conduct sensitivity analysis using calibrated parameters, as the parameters have changed remarkably from the starting predicted values. This analysis is done for several reasons, for example to identify parameters that need to be investigated more for simulating the base knowledge; to determine unimportant parameters which can be removed before doing the final modelling; to identify the parameters that contribute most and the parameters that contribute the least to the output variability (Hamby, 1994). The sensitivity of the parameters in modelling indicates the strength of the parameter (Niyonkuru et al., 2018). For our study the sensitivity of model parameters was observed with respect to the model output to classify specific parameters. Using methods used by Jewell et al., (1978) the value ranges were justified up to the maximum range of ±50% from their initial or original values (as shown in Table 4). The other parameters that were insensitive or least sensitive were kept constant to get the best fitted values.

Table 4. *The value ranges and final value used for calibration of the model (Data used: [a]Kong et al., (2017); [b]Temprano et al., (2006); [c]Huber et al., (1998), [d]Modugno et al., (2015)).*

Parameters	Initial Value	Allowed variation Range	Calibrated value	Rank of sensitivity
Impervious %	Calculated	±20%[a]	Calculated	1
Width (m)	Calculated	±30%[b]	Calculated	3
Slope%	Calculated	±30%[b]	Calculated	6
N-impervious	0.010	0.010-0.013[c]	0.012	4
N-pervious	0.13	0.01-0.32[c]	0.13	7
Dstore impervious (mm)	1.78	1.27-2.48[c]	1.27	2
Dstore pervious (mm)	3.56	2.48-5.08[c]	2.54	8
%Zero impervious	25	1-45[d]	25	5

Based on the changes in the peak or shape of the graph the rank of sensitivity were allotted to different parameters depending on the variations while doing the calibration of the model. As a result, it was identified the impervious surface affect the peak shape the most.

For the water quality part, the parameters that were sensitive in hydrological study need not to be sensitive for the water quality too. But the highest sensitive parameters identified in general are buildup functions (the maximum buildup, the buildup exponent) and wash off function (wash off coefficient and wash off exponent). Here in this study, it refers to the power buildup function and exponential wash off functions.

The water quality model was run with initial values taken from the earlier done study (Chow et al., 2012; Tu & Smith, 2018; Ball et al., 1998; Tsunokawa & Hoban, 1997). Then the model run was made by varying the quality parameters values (buildup function and wash off function) up to ±50% keeping other values same. All last the goodness fit criteria were calculated to check the efficiency of the model. The initial values used are given in Table 5.

Table 5. *Initial values from different for pollutants in different land use used for the water quality modelling (Data taken from [a]Chow et al., 2012; [b]Ball et al., 1998; [c]Tsunokawa & Hoban, 1997; [d]Li et al., 2016).*

Land use	Pollutant	Buildup C_1 (Kg/100m)	Buildup C_2 (Kg/day/100m)	Wash off D_1	Wash off D_2
Commercial	TSS	1.5[a]	1.2[a]	0.01[d]	1.8[d]
Road	TSS	1.873[b]	0.75[c]	0.012[d]	2[d]

3.7 SWMM Calibration and Validation

Since we have only two rainfall events for this study, we will use one rainfall event for calibration (PN2) and other for validation (PN1) for the stormwater quantity and quality modelling. The event that was selected for calibration is from 5th July 2023 and the event selected for validation is from 22nd and 23rd June 2023. The PN2 event lasted for 3hours and 25 minutes whereas PN1 event lasted for 5 hours 45 minutes, the total rainfall intensities were 13 and 20.3 and the peak runoff flows were 156.76 and 121.43 l/s for PN2 and PN1. The antecedent dry days for both events were 0 days. Basic information about both the events is mentioned in Table 6.

Table 6. *Details of the events selected for Calibration and Validation of the model.*

Name	Date	Duration	Peak flow (l/s)	Rainfall intensity (mm)	Antecedent dry days	Total load (kg)	Total EMC (mg/l)
Cal	5th July 2023	3h 25m	156.76	13	0	5.41	139.4
Val	22nd-23rd June 2023	5h 45m	121.43	20.3	0	9.81	351.9

The parameters used for calibration of the runoff quantity and its range in the SWMM model along with the initial values used are mentioned in the Table 4. The calibration of the model was performed several times by the adjustment of different sensitivity parameters within permissible limits (Table 4). The parameters were adjusted one at moment in such a way that the variance between the observed value and measured value were minimum. Which means if the difference between the measured value and modelled value is minimum the calibration process is completed by making sure it also fits the goodness fit criteria.

For the water quality the calibration in the model was done using the four parameters (maximum buildup, buildup coefficient, wash off coefficient, wash off exponent) for the pollutants in every land use. Each of these parameters were adjusted up to a specific range to get minimum difference between simulated and observed values in terms of event mean concentrations, peak concentrations and total pollutant load values. The main motive to do this calibration and validation is to check if the set of parameters used in the calibration can be used to simulate other storm events in the future (Niyonkuru et al., 2018). For the current study the event PN2 was calibrated and using the same parameters values the event PN1 then was validated for the runoff quantity as well as quality.

4. Results

4.1 Runoff quantity Sensitivity analysis

While doing the sensitivity analysis for the calibration of the model, the individual parameters from the Table 4 have been varied independently to understand the effect of different parameters on the model. Firstly, the sensitivity of the initial values was analysed and further the parameters were changed up to a permissible range as mentioned in the Table 4. The effect of different parameters on the peak of the flow is different. The effect of change in impervious percentage is significant and increases the runoff volume and peak when the percentage is increased. Width of the catchment mostly affects the shape of the hydrograph, by decreasing the runoff volume and increasing runoff peak with increase in the width. If there are several storm events with different intensity, then the increase in width will produce earlier and higher flow peaks, in simpler words it will show a faster response. **Slope same as width** also affects the shape of the hydrograph but is lesser sensitive, when increased it increases the runoff volume and decreases the runoff peak. Roughness of the pipes also affects the shape of the hydrograph but shows inverse effect than width and slope of the catchments. **Depression storage has very moderate effect on the hydrograph it becomes significant only when the storm event depth is low.**

Table 7. *The model performance statistics (NSE, R2, RSR and PBIAS) for the parameter variation conducted during the sensitivity analysis.*

Parameter	Change	Min	Max	Mean	Calibration			
					NSE	R^2	RSR	PBIAS
Imperviousness	-30%	0	90	46.2	0.797	0.82	0.48	1.27
	0	0	100	61.88	0.15	0.79	0.96	-40.31
	+30%	0	100	74.86	-0.55	0.62	1.28	-60.93
Width	-30%	9.22	106.97	56.79	0.17	0.74	0.95	-36.99
	0	13.71	152.81	81.13	0.15	0.79	0.96	-40.31
	+30%	17.12	198.65	105.47	0.13	0.81	0.95	-41.91
Slope	-30%	0.07	1.38	0.58	0.16	0.77	0.95	-38.68
	0	0.1	1.97	0.82	0.15	0.79	0.96	-40.31
	+30%	0.13	2.56	1.07	0.14	0.80	0.95	-41.26
N-impervious	-20%	0.008	0.008	0.008	0.14	0.81	0.95	-41.80
	0	0.01	0.01	0.01	0.15	0.79	0.96	-40.31
	+20%	0.012	0.012	0.012	0.16	0.77	0.95	-38.62
N-pervious	-10%	0.117	0.117	0.117	0.15	0.79	0.96	-40.31
	0	0.13	0.13	0.13	0.15	0.79	0.96	-40.31
	+10%	0.143	0.143	0.143	0.15	0.79	0.96	-40.31
Dstore impervious	-40%	1.27	1.27	1.27	0.116	0.83	0.99	-46.62
	0	1.78	1.78	1.78	0.15	0.79	0.96	-40.31
	+40%	2.49	2.49	2.49	0.17	0.70	0.98	-31.29
Dstore pervious	-40%	2.54	2.54	2.54	0.15	0.79	0.96	-40.39
	0	3.56	3.56	3.56	0.15	0.79	0.96	-40.31
	+40%	4.98	4.98	4.98	0.15	0.79	0.95	-40.39

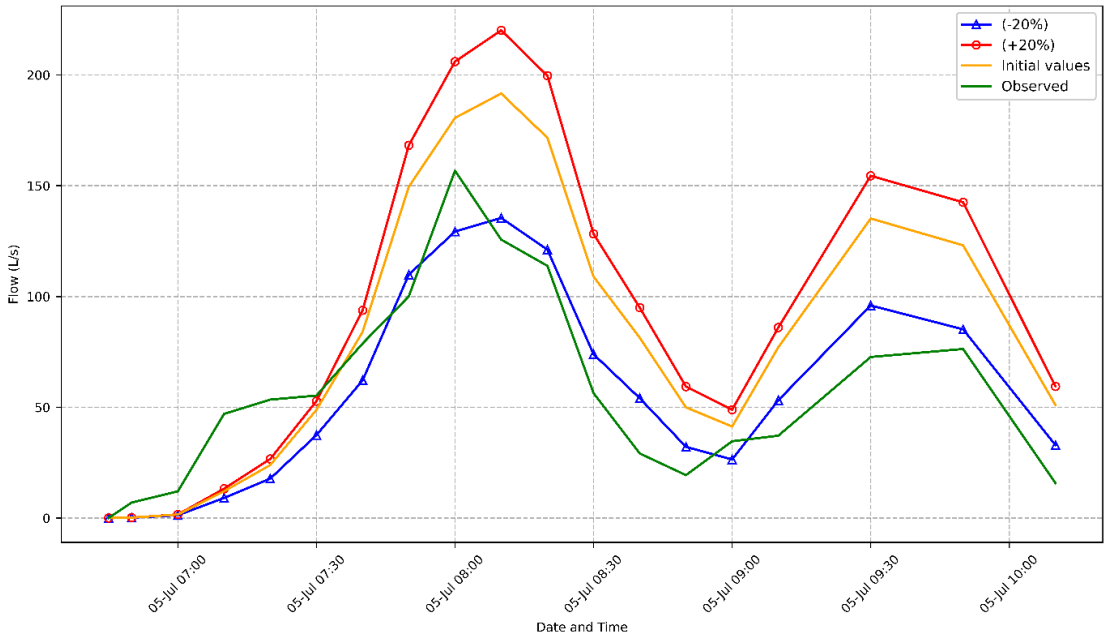

Figure 10. *The effect of perturbations before the calibration event in the sensitivity analysis in the impervious percentage for the PN2 rainfall event (±20% of initial values).*

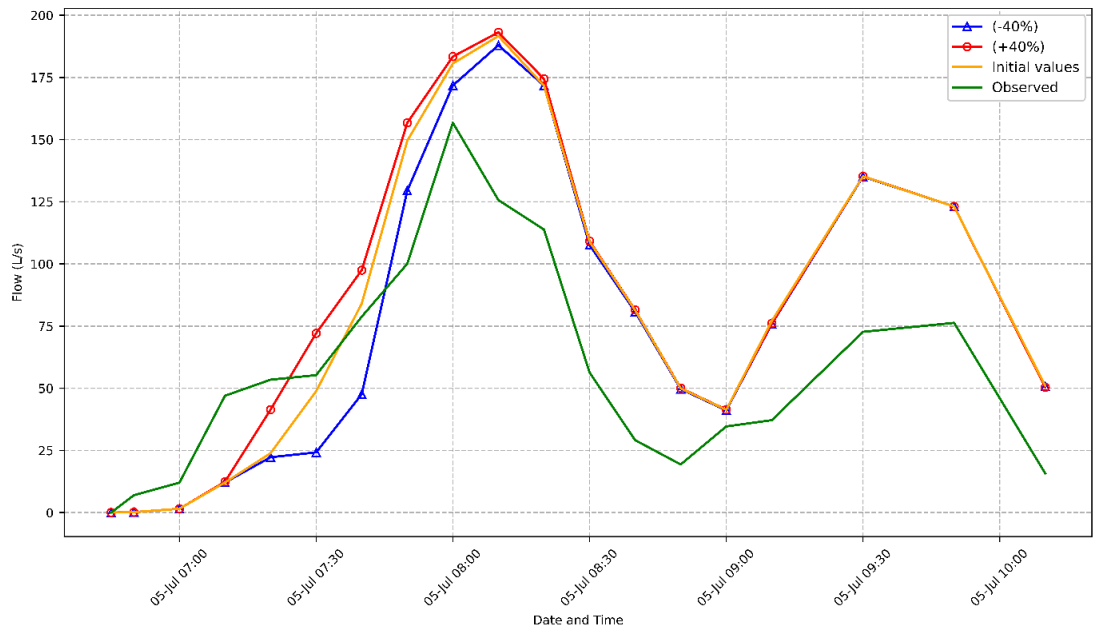

Figure 11. *The effect of perturbations before the calibration event in the sensitivity analysis in the Depth of the storage of impervious area for the PN2 rainfall event (±40% of initial values).*

As shown in Table 4 the impervious percentage is the highest and the depression storage depth of the pervious area is the lowest sensitivity parameters for the flow rate peak determination. For the calibration of the model when we varied the impervious percentage of the catchment area and the depression storage depth of pervious area with the range of -30% to +30%, the peak percentage of the runoff flow varied from -28.39% to 14.08% and 0.004% and 0.0041%, respectively. Other authors (Temprano et al., 2006; Tan et al., 2008; Chow et al.,

2012; and Maharajan et al., 2017 got almost similar results for the variation of impervious percentage.

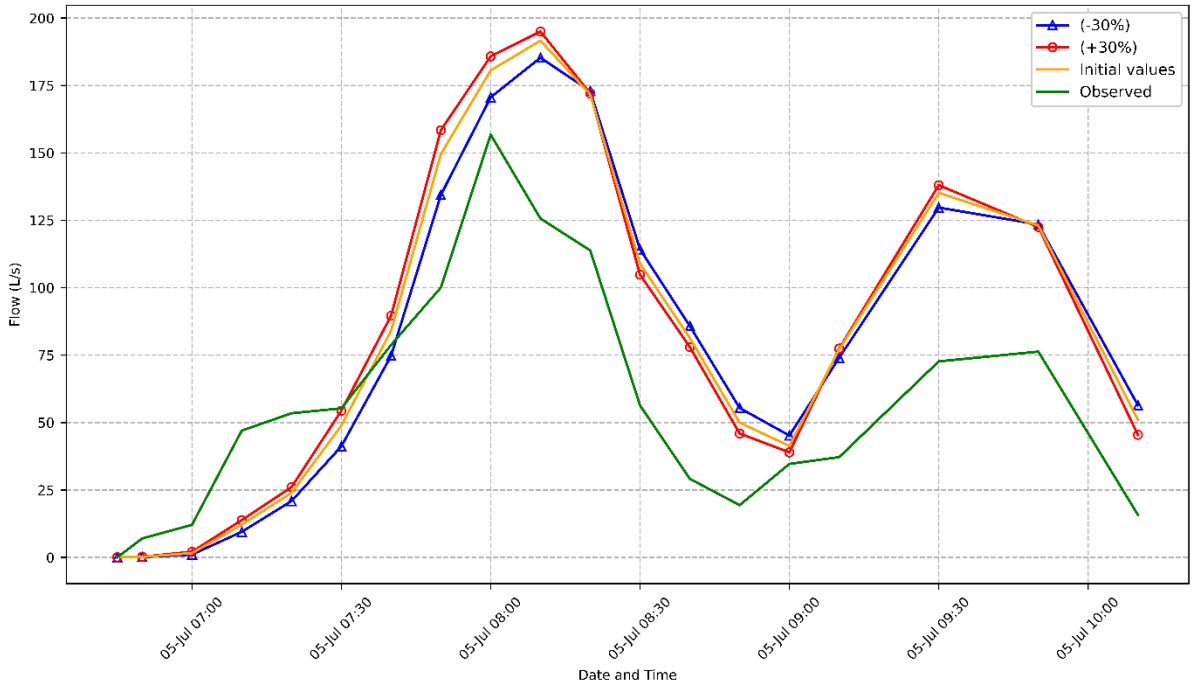

Figure 12. *The effect of perturbations before the calibration event in the sensitivity analysis in the width for the PN2 rainfall event (±30% of initial values).*

The ranking of influence of different parameters on the model are the impervious percentage on the top, followed by the depression storage depth of impervious area (Dstore-impervious), the width of the overland flow, the Manning's coefficient of the subcatchments for overland flow over the impervious area (N-Impervious), impervious area percentage with zero depression storage (%Zero-Impervious), the slope percentage, the Manning's coefficient of the subcatchments for overland flow over the pervious area (N-pervious), and lastly the depression storage depth of pervious area (Dstore-pervious), respectively. As we changed the parameters of the catchments within the maximum range of -40% to +40% we observed the changes in peak flow percentage for impervious percentage from -28.39% to 14.08% (Figure 10), for Dstore-impervious from 1.54% to -4.89% (Figure 11), for width from -5.5% to 2.89% (Figure 12), for N-impervious 2.63% to -2.75% (Figure 13), for %Zero-impervious from 0.4% to 0.5%, for slope from -2.68% to 1.39% (Figure 14), N-pervious doesn't show any variation (Figure 15) and Dstore-pervious from 0.004% to 0.004% (Figure 16), respectively. As per our result the peak flow decreases with increase in Manning's coefficient of the subcatchments for overland flow over the impervious area (N-impervious) and the depression storage depth of impervious area (Dstore-impervious) which aligns with Chow et al., 2012 findings.

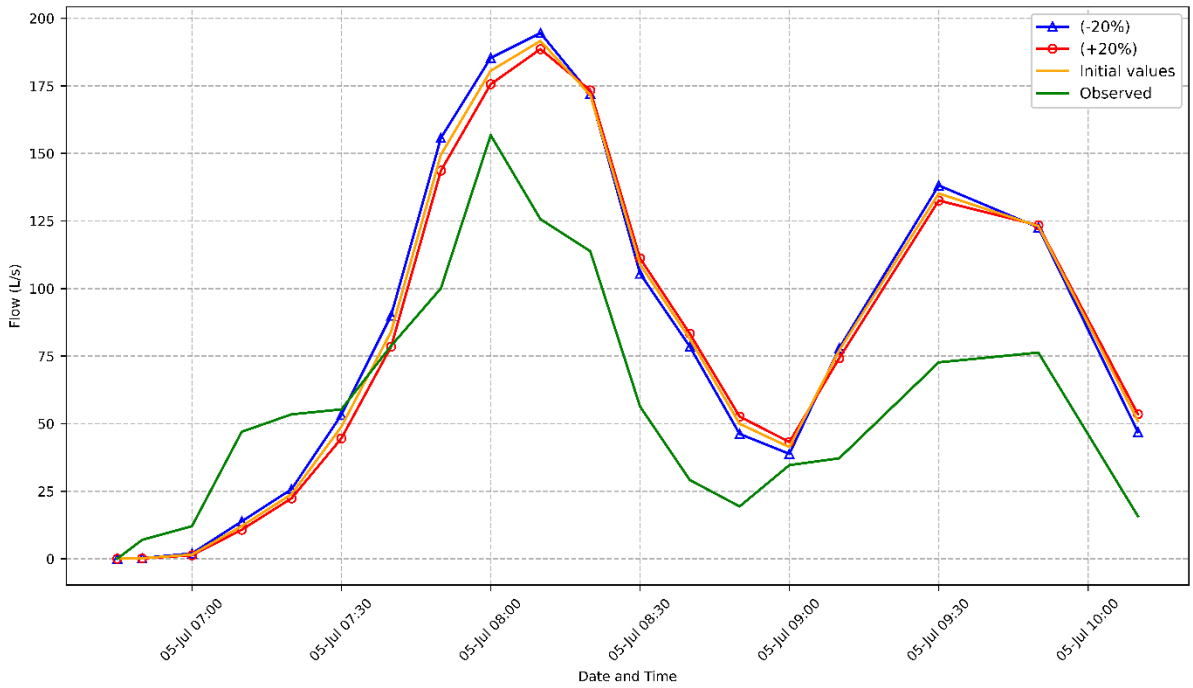

Figure 13. *The effect of perturbations before the calibration event in the sensitivity analysis in the manning's coefficient for impervious area for the PN2 rainfall event (±20% of initial values).*

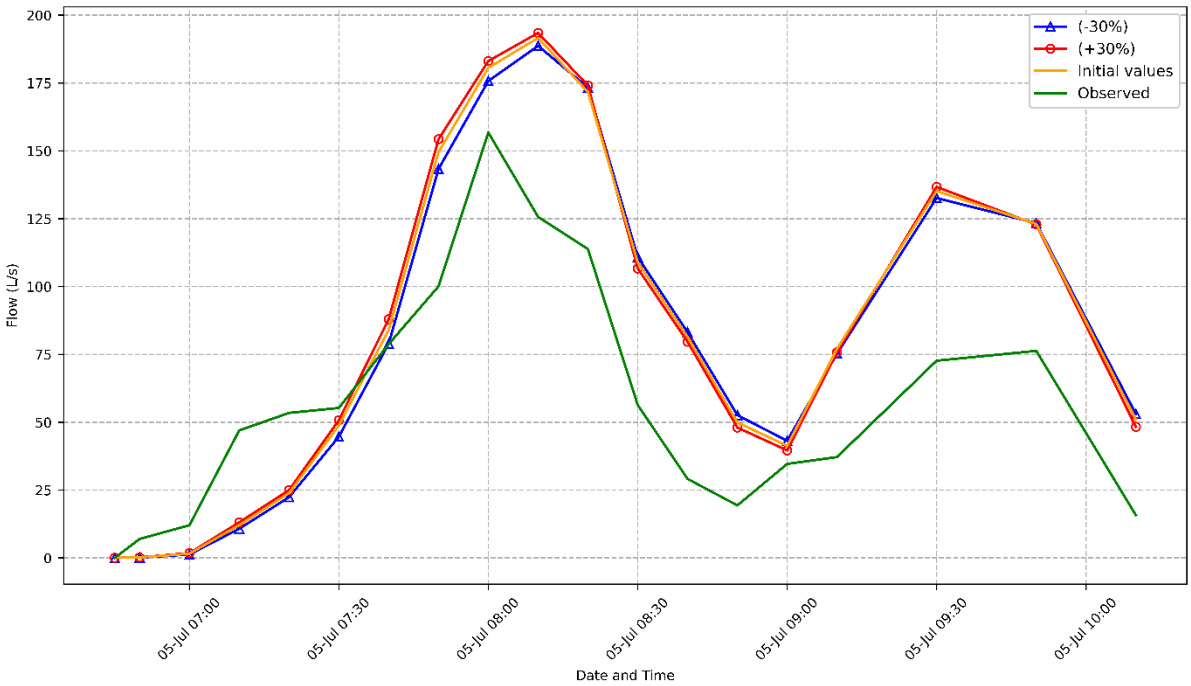

Figure 14. *The effect of perturbations before the calibration event in the sensitivity analysis in the slope for the PN2 rainfall event (±30% of initial values).*

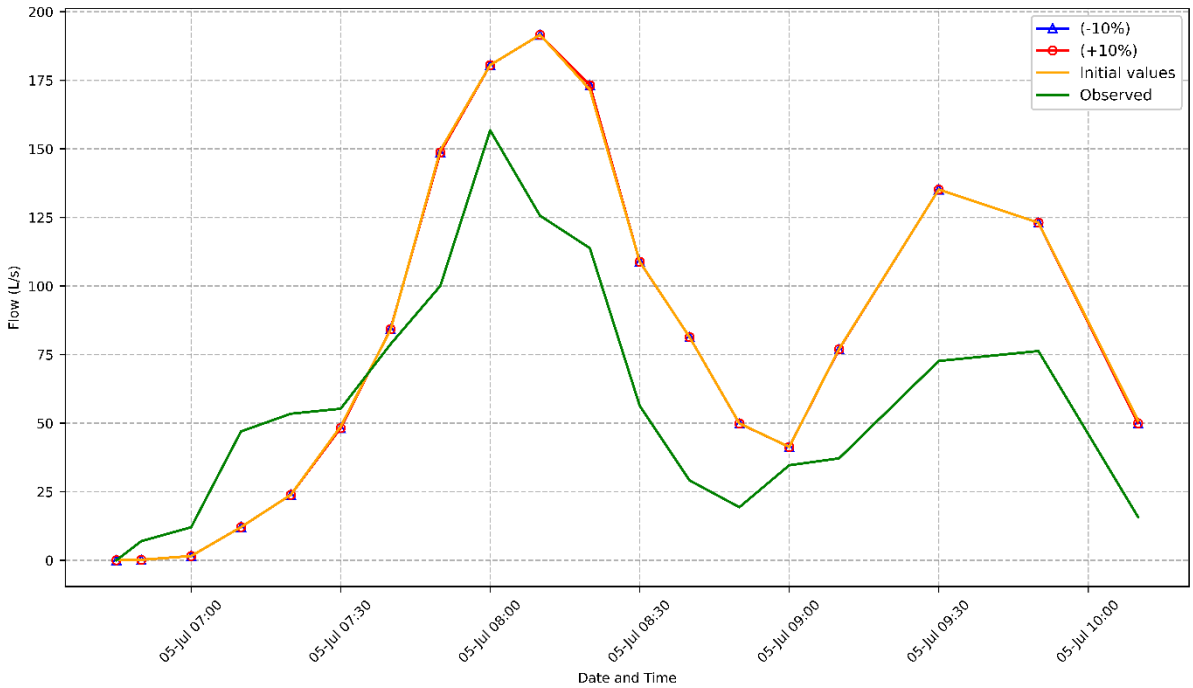

Figure 15. *The effect of perturbations before the calibration event in the sensitivity analysis in the Manning's coefficient for the pervious surface for the PN2 rainfall event (±10% of initial values).*

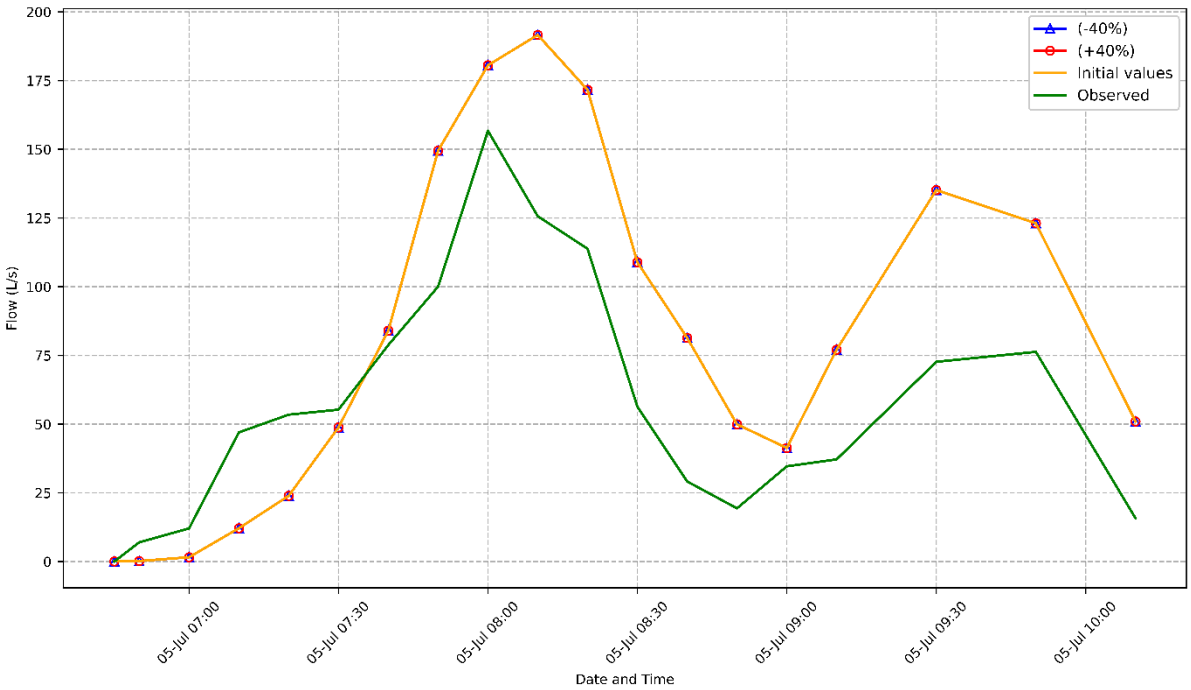

Figure 16. *The effect of perturbations before the calibration event in the sensitivity analysis in the Depth of the storage of pervious area for the PN2 rainfall event (±40% of initial values).*

Meanwhile, the flow runoff peak increased with increase in impervious percentage of the subcatchment area, the width of the flow and slope. The sensitivity analysis for our study suggests that the peak flow produced by simulation of the SWMM model was more sensitive to changes in imperviousness properties such as impervious percentage of the subcatchments

area, Manning's coefficient of the subcatchments for overland flow over the impervious area (N-impervious) and the depression storage depth of impervious area (Dstore-impervious). Also, the depression storage depth of pervious area (Dstore-pervious) showed least effect and Manning's coefficient of the subcatchments for overland flow over the pervious area (N-pervious) showed zero effect on the runoff peak, similar findings were observed by Moafi Rabori et al., 2017.

4.2 Runoff Quantity result

4.2.1 Calibration

Since we had only two sets of events as mentioned above, we choose one event for the calibration and the other for validation of the data. After running the simulations for several times and adjusting the parameters according to the sensitivity degree, further it was evaluated for the goodness of fit test using four indications as shown in the equation 7-10. During calibration the computed values of the runoff flow depicted good relationship with the observed value. Figure 17 depicts nice comparison between the computed and observed runoff flow data for the rainfall event. The figure depicts good fit between the observed and computed data which was also confirmed by the goodness of fit test. The goodness-of-fit test results for the calibration and validation are shown in Table 8. For the calibration event, the value of NSC was 0.83 which is greater than 0.65, R^2 was 0.84 which is greater than 0.8, RSR was 0.38 which is less than 0.55, and PBIAS (%) was 6.22% which is less than +15%. All these values suggest that the model is well calibrated according to Moriasi et al., 2007.

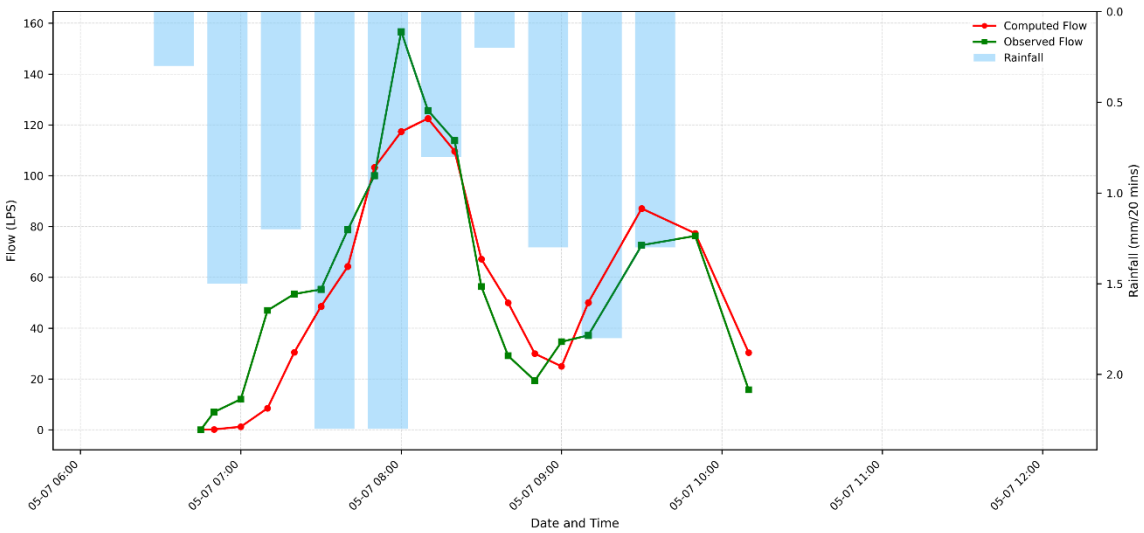

Figure 17. *Comparison between observed flow hydrograph (green line) and computed flow hydrograph (red line) and rainfall bar graph (light blue colour) for Calibration PN2 rainfall event (5th July 2023).*

4.2.2 Validation

The database of the adjusted parameters that we got by calibrating the SWMM model of PN2 rainfall event was used to validate PN1 rainfall event without making any additional changes in the database. When the simulation was run a satisfactory alignment between the computed and observed runoff was seen. The hydrograph of this event passed the goodness fit test between the computed and observed runoff discharge time series, which is depicted in Figure 18. The goodness fit test of the validation events (Table 8) results in NSC value of 0.52,

R^2 value of 0.67, RSR value of 0.58 and PBIAS value of -8.6%, respectively which is a satisfactory result and in acceptable range (Moriasi et al., 2007).

Table 8. *Goodness-of-fit test results for evaluating the dependability of SWMM model calibration and validation phases for runoff quantity simulation (G*: Good and S**: Satisfactory).*

Modelling Phase	Event Date	Goodness fit test	Performance rating
Calibration	5th July 2023	NSC R^2 RSR PBIAS (%)	0.84(G*) 0.85(G) 0.38(G) 6.23(G)
Validation	22nd and 23rd June 2023	NSC R^2 RSR PBIAS (%)	0.52(S**) 0.67(S) 0.58(S) -8.65(G)

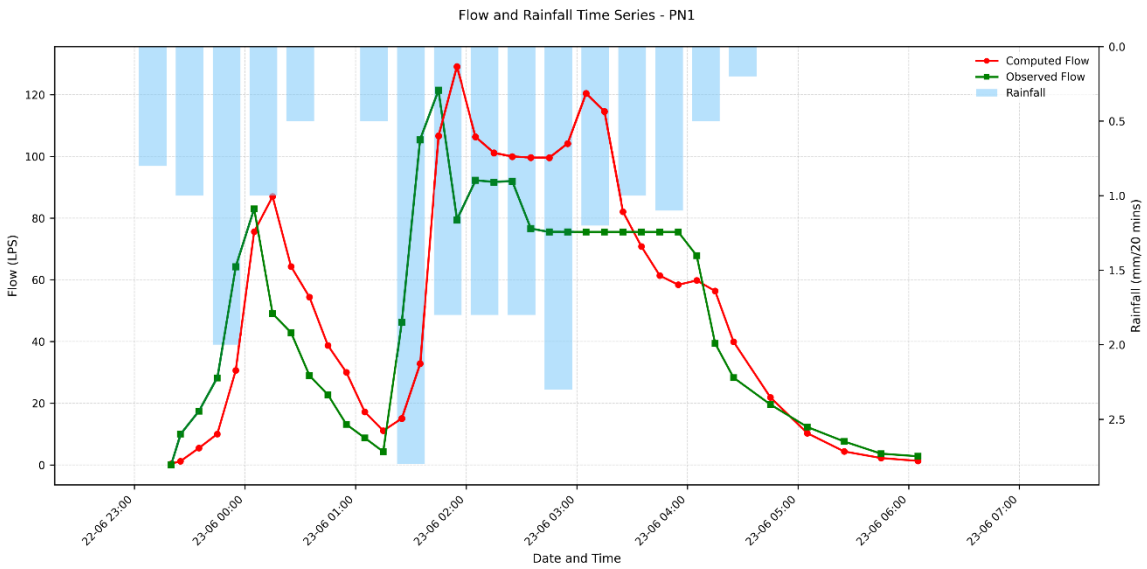

Figure 18. *Comparison between observed flow hydrograph (green line) and computed flow hydrograph (red line) and rainfall bar graph (light blue colour) for Validation PN1 rainfall event (22nd and 23rd June 2023)*

4.3 Runoff quality Sensitivity analysis

While analysing Total suspended solids variations it was observed that maximum buildup effected the amount of pollutant the most. When the MB was deviated to a change of ±50% the peak loads were changed (Figure 19). The percentage of deviation caused same percentage of change in the concentration which further also affects the total load. Which implies that the fraction of pollutant washed off by stormwater will be same, but when the maximum buildup is altered it changes the amount of pollutants. Similar observations were reported by Borris et al. (2013). Out of all four parameters highest sensitivity was displayed by the maximum buildup for the calibration event. The change in the value of MB of +50% shows an effect of change in relative error by -20.98%, whereas the change in the value of MB of -50% shows an effect of change in relative error by +59.63%.

Table 9. *The model performance statistics (NSE, R2, RSR and PBIAS) for the parameter variation conducted during the sensitivity analysis after calibration.*

Parameter	Change	Commer. Value	Road Value	Calibration			
				NSE	R^2	RSR	PBIAS
MB	-50%	0.75	0.7	-0.60	0.71	1.14	49.28
	0	1.5	1.4	0.70	0.71	0.49	-0.48
	+50%	2.25	2.1	-0.49	0.71	1.16	-50.24
BE	-50%	0.6	0.55	0.70	0.71	0.49	-0.48
	0	1.2	1.1	0.70	0.71	0.49	-0.48
	+50%	1.8	1.65	0.70	0.71	0.49	-0.48
WC	-50%	0.0275	0.002	-0.03	0.73	0.89	-38.59
	0	0.055	0.004	0.70	0.71	0.49	-0.48
	+50%	0.0825	0.006	-0.41	0.44	0.97	41.75
WE	-50%	0.35	2.25	-4.6	0.40	1.63	-30.23
	0	0.7	4.5	0.70	0.71	0.49	-0.48
	+50%	1.05	6.75	-0.66	0.073	1.08	3.97

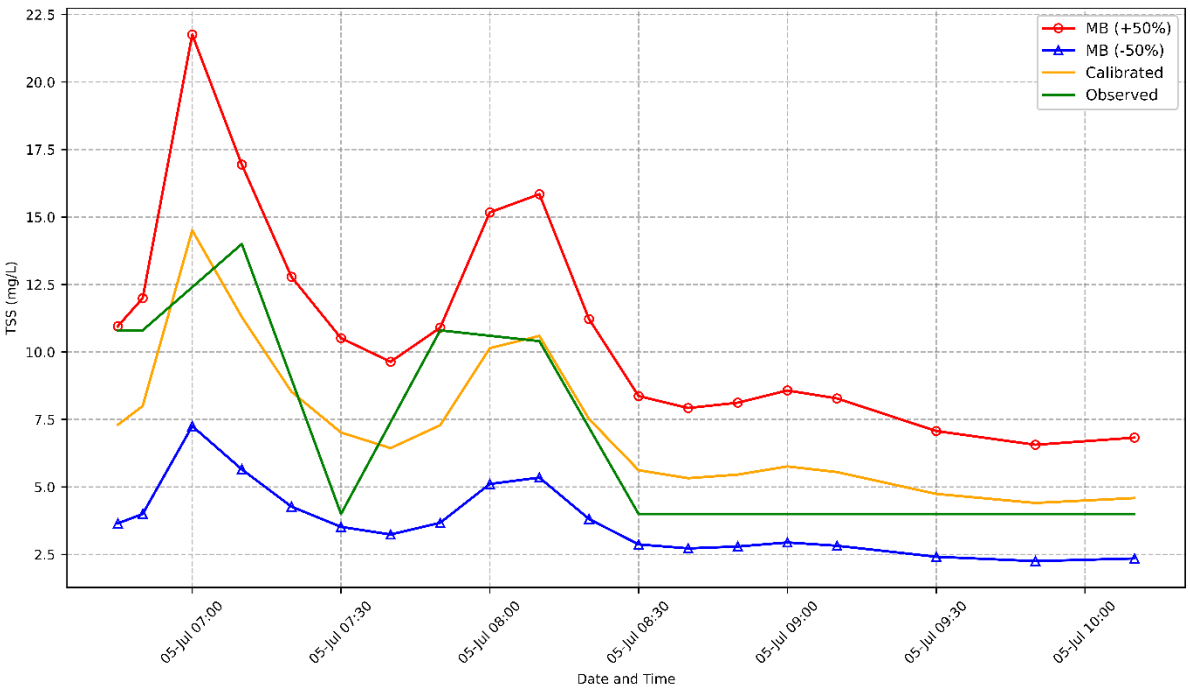

Figure 19. *The effect of perturbations in the maximum build-up after calibration event for sensitivity analysis (±50% of calibrated values).*

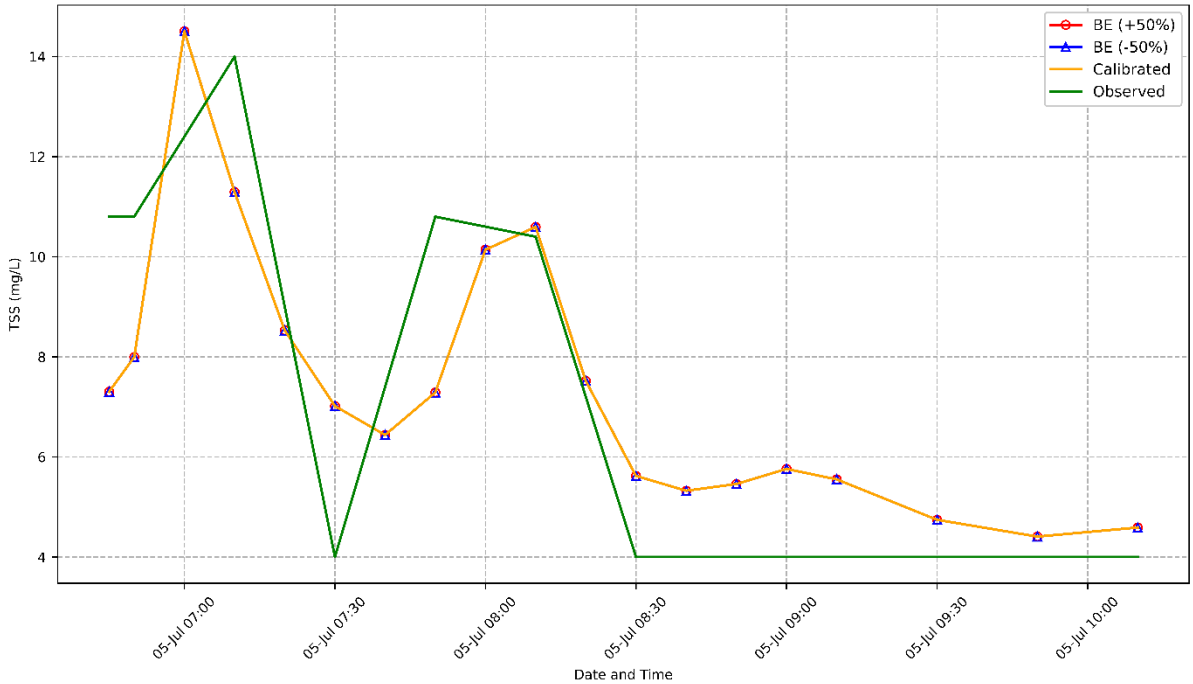

Figure 20. *The effect of perturbations in the build-up exponent after calibration event for sensitivity analysis (±50% of calibrated values).*

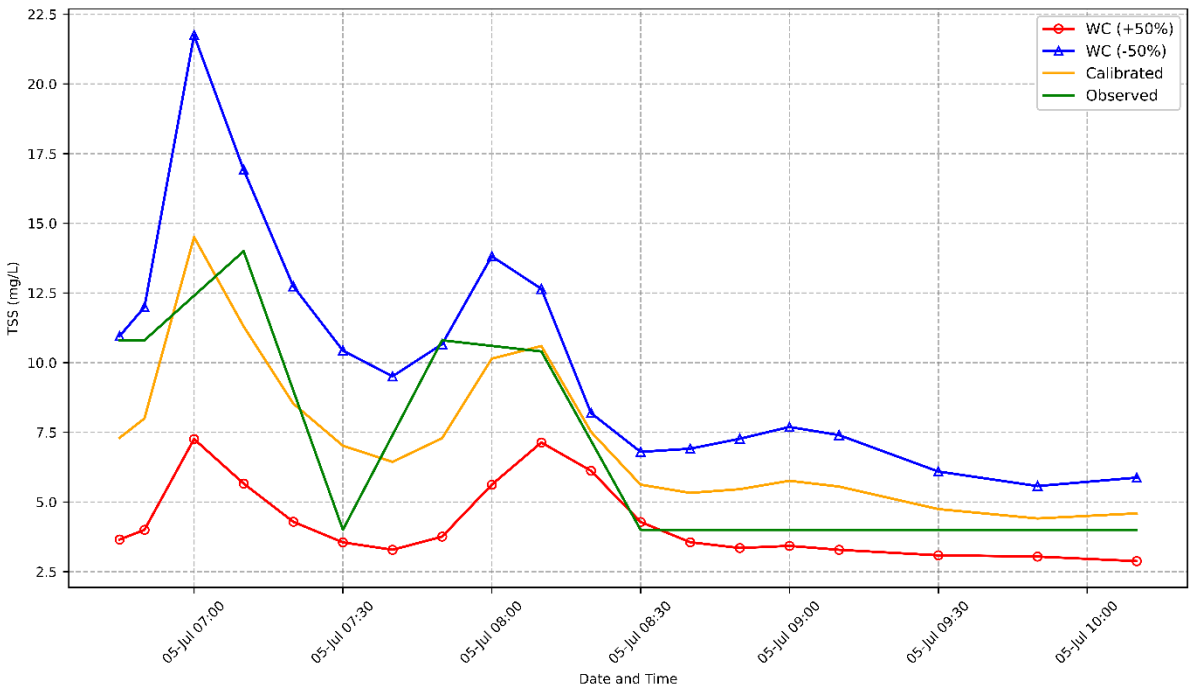

Figure 21. *The effect of perturbations in the washoff coefficient after calibration event for sensitivity analysis (±50% of calibrated values).*

How fast the maximum buildup is reached is determined by the buildup exponent. The least sensitive quality parameter in this model is the buildup exponent (Figure 20). If the BE value given to the model is small, required peak is not attained depending on the conditions of the rest of the quality parameters. The higher the BE value increases the better fit between the computed and observed data can be observed. The reason behind this variation is the rapid

accumulation of the pollutants that are presented by high BE. The faster accumulation results in large time between two peaks so that more accumulation of particulate matter can take place and finally at the end of high intensity rainfall event it can be washed off. But if the value of BE is small then it doesn't get enough time to accumulate particulate matter and hence the pollutant in wash off is lesser.

Another sensitive quality parameter after the maximum buildup is the washoff coefficient. When the WC was deviated to a change of ±50% the peak loads were changed up to -50% and +49.7% (Figure 21). The increase in the percentage of deviation causes increase the fraction of pollutant washed off by stormwater. Therefore, if the MB or WC increases, the concentration increases. But the only difference is that the MB rises the quantity of surface pollutants, when there are more pollutants available for wash-off. Whereas the increase in WC rises the efficacy of washoff, which implies the release of more pollutant due to runoff. The change in the value of MB of +50% shows an effect of change in relative error by +59.58%, whereas the change in the value of MB of -50% shows an effect of change in relative error by -20.84%.

During a rainfall event the washoff pattern is highly impacted by the washoff exponent. As it is shown in Equation 6 the WE is dependent on the stormwater runoff flow, which implies that the concentration of the pollutants during washoff is also dependent on the stormwater flow. But this concentration can also vary depending on the value of WE (Figure 22). When the WC was deviated to a change of ±50% the peak loads were changed up to -36.95% and +75.52%. The change in the value of MB of +50% shows an effect of change in relative error by +49.14%, whereas the change in the value of MB of -50% shows an effect of change in relative error by -41.60%.

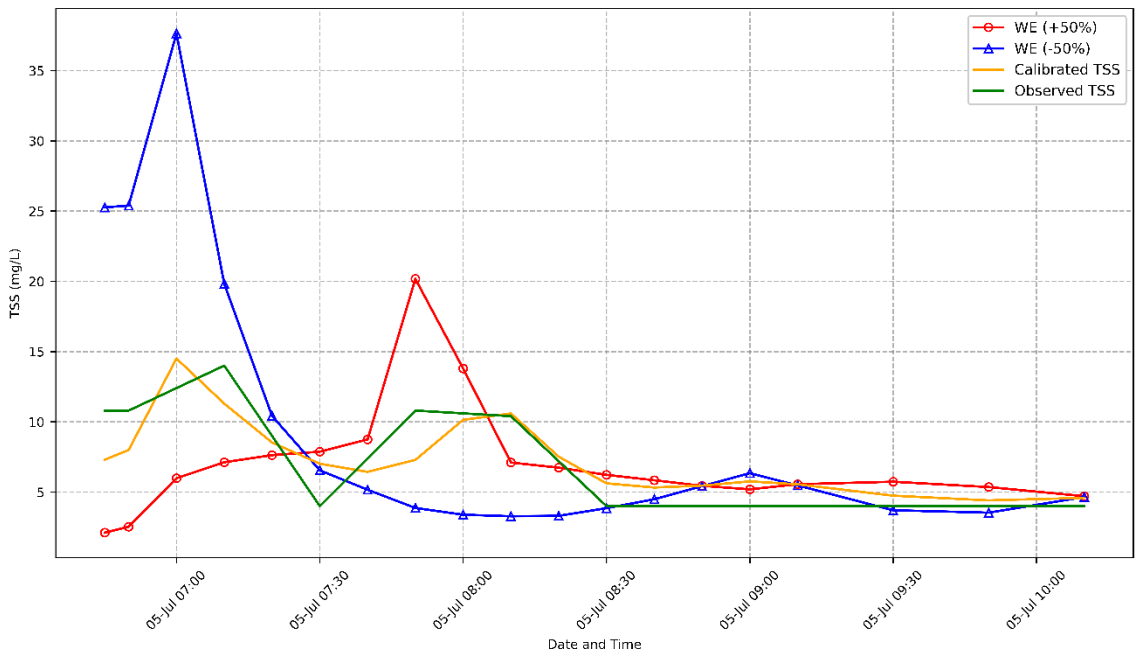

Figure 22. *The effect of perturbations in the washoff exponent after calibration event for sensitivity analysis (±50% of calibrated values).*

4.4 Runoff quality result

4.4.1 Calibration data

After running the simulation several time when the goodness of fit was achieved for TSS pollutant for the PN2 event the best fitted value for the quality parameters were attained. Table 10 represents the result of calibrated data in every land use for the buildup and washoff parameters for the quality modelling. During calibration the computed values of the runoff TSS quality depicted good relationship with the observed value. Figure 21 depicts nice comparison between the computed and observed TSS quality data for the rainfall runoff event. The figure depicts good fit between the observed and computed data which was also confirmed by the goodness of fit test. The goodness-of-fit test results for the calibration and validation are shown in Table 11. For the calibration event, the value of NSC was 0.70 which is greater than 0.65, R^2 was 0.71 which is greater than 0.6, RSR was 0.49 which is less than 0.55, and PBIAS (%) was -0.48% which is less than -15%. All these values suggest that the model is well calibrated according to Moriasi et al., 2007.

Table 10. *Best buildup and washoff input parameters for the water quality module in the SWMM model*

Land use	**Pollutant**	**Buildup C_1 (Kg/100m)**	**Buildup C_2 (Kg/day/100m)**	**Wash off D_1**	**Wash off D_2**
Commercial	TSS	1.5	1.2	0.055	0.7
Road	TSS	1.4	1.1	0.004	4.5

4.4.2 Validation

The database of the adjusted parameters that we got by calibrating the SWMM model of PN2 rainfall event for the runoff quality was used to validate PN1 rainfall event without making any additional changes in the database. When the simulation was run a satisfactory alignment between the computed and observed runoff was seen (Figure 23 and 24). The pollutograph of this event couldn't pass the goodness fit test between the computed and observed runoff discharge time series, which is depicted in Figure 5. The goodness fit test of the validation events results in NSC value of 0.30, R^2 value of 0.68, RSR value of 0.62 and PBIAS value of 30.68%, respectively which is not a satisfactory result and isn't in acceptable range (Moriasi et al., 2007).

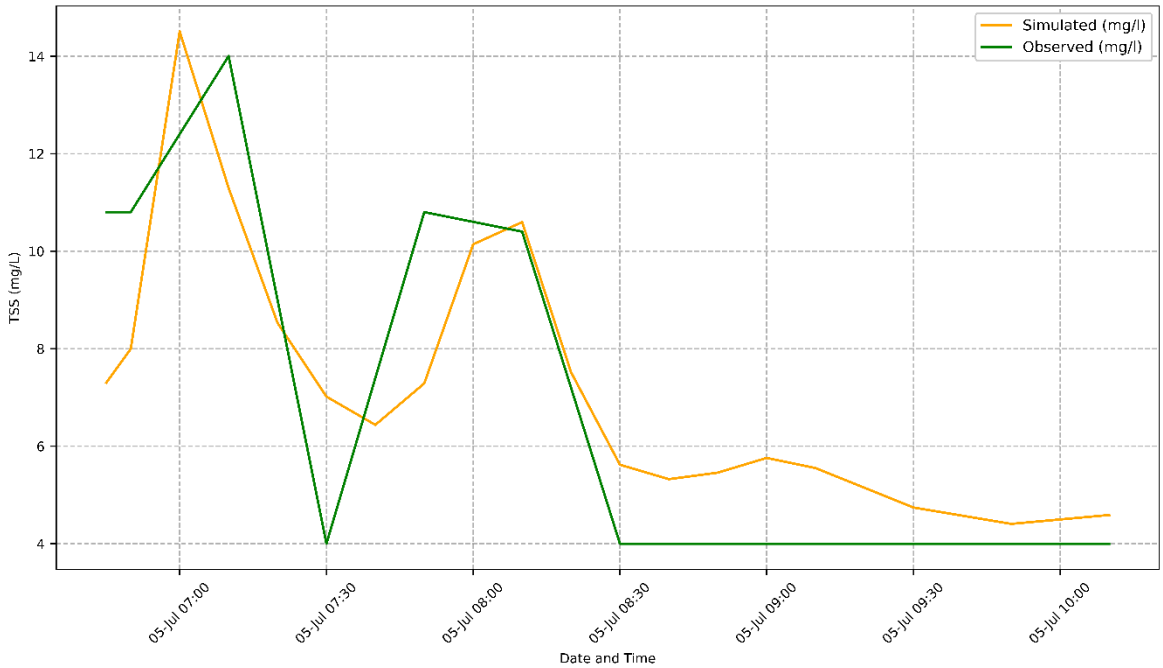

Figure 23. *Comparison between observed data (green line) and simulated data (yellow line) for Calibration PN2 rainfall event (05th July 2023)*

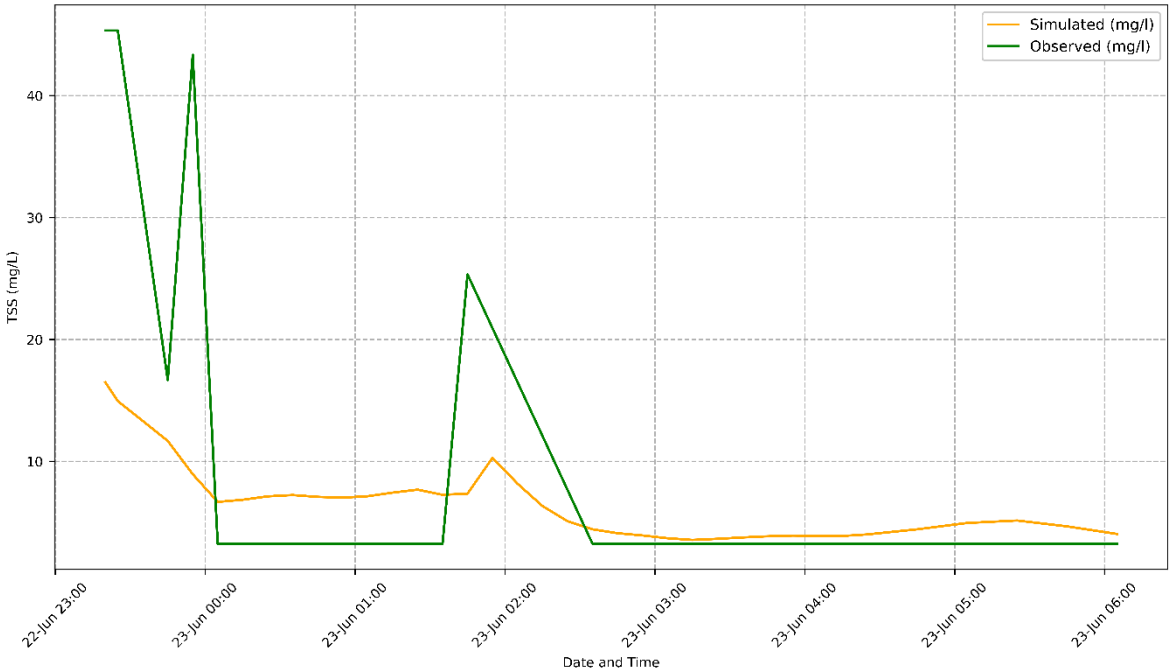

Figure 24. *Comparison between observed data (green line) and simulated data (yellow line) for Validation PN1 rainfall event (22nd and 23rd June 2023).*

Another fitness test was conducted by making a plot of simulated vs observed data of TSS concentration (Figure 25 and 26). A linear regression line (simulated value =observed value) was also formed in this figure; its slope was matched with perfect match value (1:1 value). The equation y=mx+c was also added in the figure, where m is the value of the under prediction (slope value lesser than 1) or over prediction (slope value greater than 1) of the simulated model compared to the measured value.

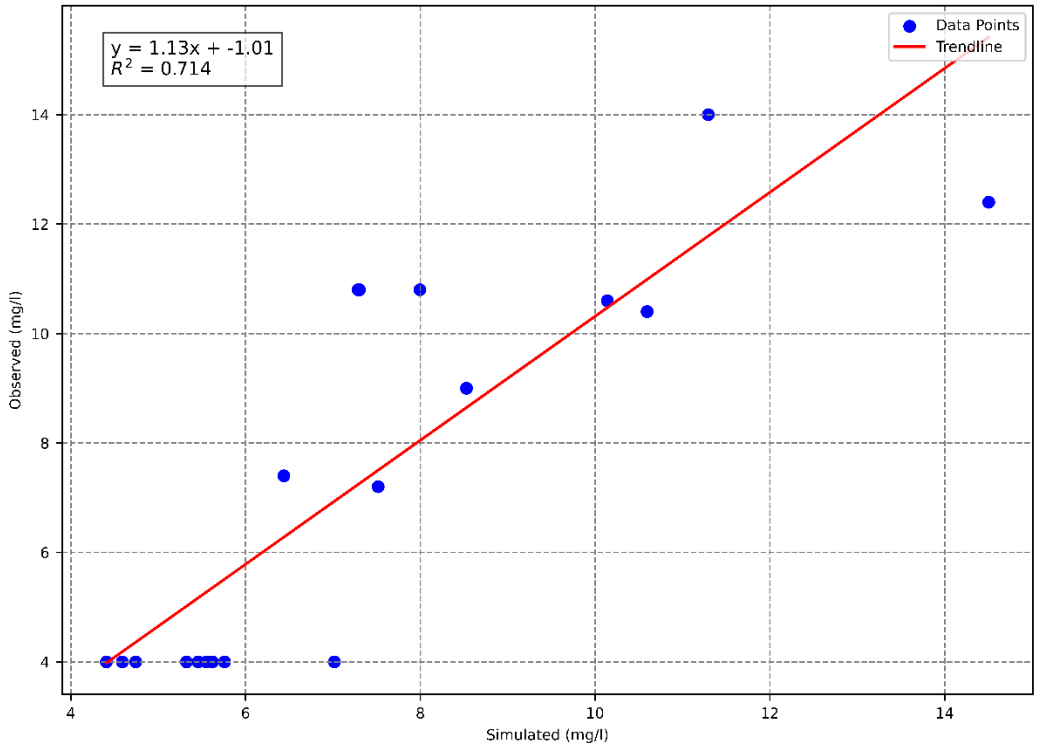

Figure 25. *Correlation between observed and simulated concentrations of the pollutants for the Calibration rainfall event PN2 (5th July 2023).*

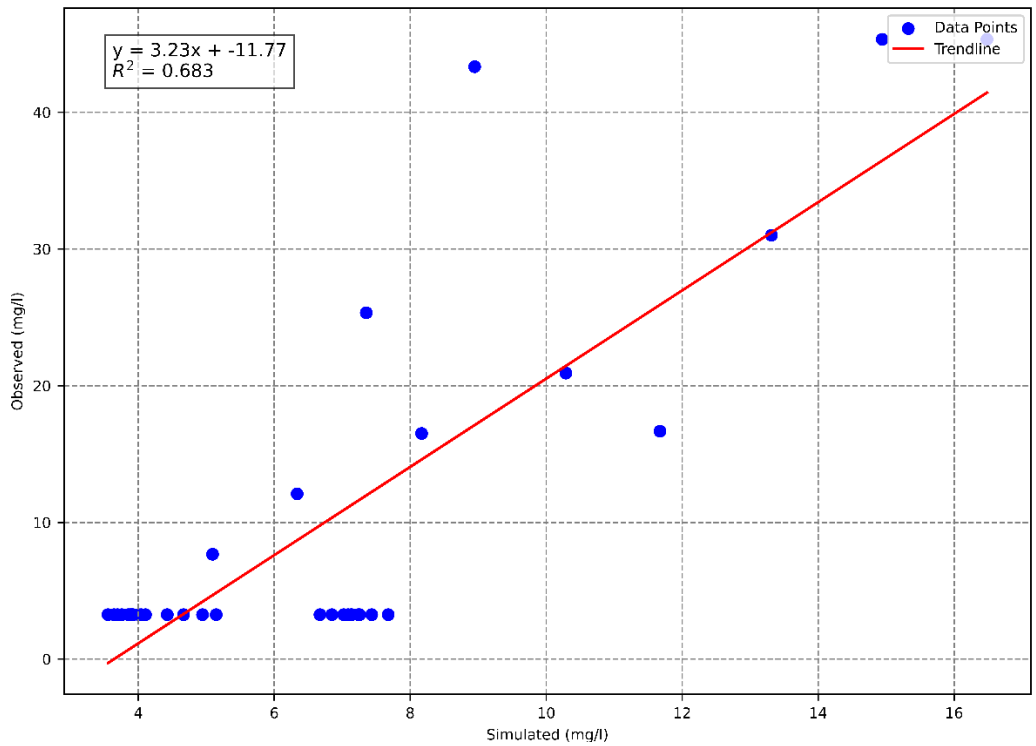

Figure 26. *Correlation between observed and simulated concentrations of the pollutants for the Validation rainfall event PN1 (22nd and 23rd June 2023).*

Along with that, the square of the correlation of coefficient (R^2) of the regression line is shown in the figure. If the value of R^2 and m is close to 1 then it is best fitted (Tsihrintzis and

Hamid, 1998). In our study, the calibrated event shows the value for R^2 and m of 0.714 and 1.13, whereas the validated event shows the value for R^2 and m of 0.683 and 3.23. This indicates that the validation gives unsatisfactory value during the simulation of water quality modelling. In both calibration (Figure 25) and validation (Figure 26), the model fits the data well ($R^2 = 0.71$ and 0.68). At the lower concentration range, however, some points do not follow the trendline. This happens because pollutant build-up and wash-off are harder to model at low levels, and outlet data average out the variable behaviour of the catchment. Small errors from rainfall or TSS measurements at low values can also add to the mismatch. Such deviations are common in stormwater quality modelling.

Table 11. *Goodness-of-fit test results for evaluating the dependability of SWMM model calibration and validation phases for runoff quality (TSS) simulation (G*: Good, S**: Satisfactory and US#: Unsatisfactory).*

Modelling Phase	Event Date	Goodness fit test	Performance rating
Calibration	5th July 2023	NSC R^2 RSR PBIAS (%)	0.70(G*) 0.71(S**) 0.49(G) -0.48(G)
Validation	22nd and 23rd June 2023	NSC R^2 RSR PBIAS (%)	0.30(US#) 0.68(S) 0.63(S) 30.67(US)

5. Discussion

Unlike most existing studies, which apply SWMM to large urban areas or city-wide scales, this study applied the model to a small commercial/light industrial catchment. This study helped in developing a calibrated SWMM model for the simulation of stormwater runoff quantity and quality for a small urban catchment in Bargteheide, Germany. This scale-specific application allowed for a detailed calibration and validation using locally observed rainfall, flow, and water quality data. The approach demonstrated how a model designed for larger urban areas can be effectively adapted to a small-scale catchment, highlighting both the opportunities and the challenges in model calibration and pollutant simulation at such scales. The discussion of this study includes the practical findings and the literature, also identifies the critical modelling and monitoring issues which influenced the model performance.

5.1 Performance of the water quantity model

The SWMM model calibration for runoff quantity showed good agreement with observed data (NSC = 0.84, R^2 = 0.85), confirming that the hydrological parameters were well-represented for the selected rainfall event. The validation (NSC = 0.52, R^2 = 0.67) indicated satisfactory performance, though with noticeable discrepancies. These findings are consistent with literature benchmarks for urban catchment hydrology modeling (Moriasi et al., 2007).

From sensitivity analysis, impervious percentage was identified as the most sensitive parameter, significantly influencing the peak flow and hydrograph shape. This aligns with findings by Temprano et al. (2006) and Chow et al. (2012), who emphasize that imperviousness directly affects runoff generation and timing. The depression storage and width also showed high sensitivity, particularly for low-intensity storms, further supporting previous observations that minor changes in surface retention can have a disproportionate effect in small urban catchments (Rossman, 2010). While Manning's roughness coefficients and slope showed moderate sensitivity, pervious area parameters like Dstore-pervious had negligible influence—confirming that impervious surfaces dominate the runoff response in commercial catchments (Moafi Rabori et al., 2017). These sensitivities validate the parameter choices for calibration and demonstrate the appropriateness of SWMM in modeling such environments.

5.2 Performance of the water quality model

The calibration for TSS loads using power buildup and exponential washoff functions yielded strong model performance (NSC = 0.70, R^2 = 0.71, RSR = 0.49, PBIAS = -0.48%). However, the validation results showed poor fit (NSC = 0.30, R^2 = 0.68, RSR = 0.63, PBIAS = 30.68%), suggesting low transferability of the calibrated parameters across events. This is a common challenge in water quality modeling, where factors such as rainfall intensity, antecedent dry period (ADD), and catchment behavior vary greatly between events (Egodawatta et al., 2009).

Sensitivity analysis identified maximum buildup (C1) as the most influential parameter for pollutant load. Adjusting this parameter by ±50% led to changes in model error ranging from -20.98% to +59.63%, emphasizing its critical role. Similarly, the washoff coefficient (D1) greatly affected pollutant mobilization, with increased values leading to higher simulated TSS concentrations. These results are consistent with the literature, where buildup quantity and washoff efficiency are considered key drivers of stormwater quality response (Borris et al., 2013; Miguntanna et al., 2013). Remarkably, buildup exponent (C3) showed minimal

sensitivity, suggesting that accumulation rate plays a secondary role compared to the total pollutant mass available for wash-off. This finding suggests that for short, high-intensity rainfall events, pollutant mobilization is governed more by washoff dynamics than accumulation kinetics.

This result is logical for the Bargteheide catchment, which is dominated by impervious commercial and road surfaces. The maximum buildup parameter (C1) represents the total pollutant mass that can accumulate on these surfaces, and therefore strongly controls pollutant loads once rainfall occurs. The high sensitivity of the washoff coefficient (D1) also reflects the efficient mobilization of solids on smooth, hydraulically connected surfaces. By contrast, the low sensitivity of the buildup exponent (C3) is consistent with the short antecedent dry periods in both monitored events, where pollutant stocks had not yet reached saturation.

5.3 Integration of Model and monitoring data

In this study, modelling of stormwater quantity and quality data were limited to two rainfall events. But there could be more limitations in this study. For instance, representing catchment-wide behavior using a single outlet sampling point introduces spatial uncertainty, particularly in heterogeneous urban environments (Gandin, 1988; (Fletcher and Deletic, 2008).

The assumption of zero antecedent dry days (ADD) simplified the model setup but likely underestimated the pollutant mass available at the onset of rainfall. While mass balance principles still apply, SWMM represents buildup through functions that use ADD as a proxy for pollutant accumulation. By fixing ADD to zero, the model assumed minimal surface stock prior to each event, which is inconsistent with field evidence showing that even short dry periods allow substantial deposition, with up to 80% of buildup occurring in the first week (Egodawatta et al., 2009). Consequently, the wash-off process was modelled with an artificially low initial pollutant pool, explaining the underestimation of simulated TSS loads relative to observations.

Furthermore, rainfall heterogeneity and catchment-specific infiltration conditions introduced additional variability. The June 22–23 event had higher rainfall intensity but resulted in poorer model performance, likely due to more complex hydrodynamic behaviour or sensor uncertainties during peak flow. A key limitation was that rainfall data were obtained from a single gauge, which may not have captured local spatial variability in storm intensity and duration, even though pollutant wash-off is highly sensitive to such variability. Similarly, flow measurements were available only at the basin inlet, meaning that upstream processes such as local infiltration, backwater effects, or surcharging within the pipe network could not be represented in the model. In terms of water quality, the limited sampling frequency may have missed peak TSS concentrations during the first flush, constraining calibration of buildup and wash-off parameters. These limitations highlight that improved model performance requires more spatially distributed rainfall monitoring, multiple flow measurement points, and higher-frequency or continuous TSS sampling to better capture the dynamics driving stormwater runoff and pollutant transport in small urban catchments.

5.4 Challenges in Calibration and Validation

The rapid change in model performance during water quality modelling from calibration to validation indicates that event-specific calibration is essential. Calibration results for a single event may not be enough to generalize across varied hydrometeorological conditions. The

challenge is intensified in water quality modelling where pollutant sources, antecedent conditions, and rainfall intensity interact non-linearly, producing highly variable responses.

Another major factor that was observed and could be the reason behind this unsatisfactory validation of the water quality TSS simulation is the Antecedent dry days. For the rainfall event PN2 we don't have any antecedent dry days but for the rainfall event PN1 it is bit confusing (Figure 4 and 5). As displayed in figure 4 before the rainfall event days almost every day there were no rainfall but right before the event there was a high intensity rainfall event. This thing may lead to rise in the initial buildup for which we require prior knowledge as the software cannot model it. Since theoretically there are no antecedent days right before the event we cannot input the dry days values in the model. The only way to match the simulated TSS values and observed TSS values for the PN1 data is increasing the maximum buildup but that might create problem in validating the other event.

In terms of parameter uncertainty, the limited sample size restricts the ability to conduct robust statistical calibration. This calls for future work involving long-term monitoring, multi-event calibration, and potentially machine learning techniques to improve generalizability.

6. Conclusion

This study demonstrates the applicability of the SWMM model to simulate and analyze both the hydrological response and water quality dynamics of stormwater runoff in a small urban catchment in Germany. The model was effectively calibrated using a storm event, achieving high accuracy for runoff flow and a good fit for TSS concentrations. The model's sensitivity analysis identified impervious surface percentage and depression storage as the most critical parameters for flow simulation, while maximum buildup and washoff coefficient were the most influential for pollutant load predictions.

However, model validation for TSS quality revealed performance limitations, indicating that a single calibration event may not sufficiently represent all hydrological conditions, especially when antecedent dry days or rainfall variability are involved. This outcome stresses the importance of multi-event calibration and the inclusion of more precise pollutant sources such as atmospheric deposition, baseflow, and detailed land use classification.

Despite these limitations, the study provides a foundational understanding of the processes governing pollutant transport in urban environments and demonstrates the practicality of SWMM for small-scale water quality studies. Future research should focus on integrating additional data sources (such as independent infiltration/groundwater data, continuous TSS monitoring, initial buildup etc) and expanding the model to include more pollutant types and longer monitoring periods to improve robustness and predictive capability.

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List of Appendices

Appendix 1. Raw data for the parameters of subcatchments calculated using aerial photograph and the manual.

Name	Raingage	Outlet	Area	% impervio	Width	% slope	N-imperv	N-perv	S-imperv	S-pervious	Pct Zero	Route to	Pct Routed
C1	RG1	133705	0.92	56	64.542	0.11	0.01	0.13	1.778	3.56	25	impervious	100
C2	RG1	133707	0.58	68.96	70.629	0.2	0.01	0.13	1.778	3.56	25	impervious	100
C3	RG1	133747	0.47	0	70.82	0.14	0.01	0.13	1.778	3.56	25	impervious	100
C4	RG1	133709	0.56	55.36	78.672	1.48	0.01	0.13	1.778	3.56	25	impervious	100
C5	RG1	133749	0.77	31.43	130.318	1.82	0.01	0.13	1.778	3.56	25	impervious	100
C6	RG1	133731	0.58	91.37	94.5	0.84	0.01	0.13	1.778	3.56	25	impervious	100
C7	RG1	133745	0.61	60.65	83.987	0.84	0.01	0.13	1.778	3.56	25	impervious	100
C9	RG1	133741	0.54	35.19	77.84	1.93	0.01	0.13	1.778	3.56	25	impervious	100
C10	RG1	133741	0.66	56.06	78.22	0.55	0.01	0.13	1.778	3.56	25	impervious	100
C11	RG1	133737	1.13	80.53	89.21	1.76	0.01	0.13	1.778	3.56	25	impervious	100
C12	RG1	133753	0.57	0	57.94	1	0.01	0.13	1.778	3.56	25	impervious	100
C13	RG1	133757	1.28	60.16	115.53	1.3	0.01	0.13	1.778	3.56	25	impervious	100
C14	RG1	133763	0.45	37.78	59.17	0.1	0.01	0.13	1.778	3.56	25	impervious	100
C15	RG1	133763	0.58	55.17	88.53	0.65	0.01	0.13	1.778	3.56	25	impervious	100
C16	RG1	133759	0.79	78.48	112.97	0.34	0.01	0.13	1.778	3.56	25	impervious	100
C17	RG1	133751	0.74	81.08	107.06	0.85	0.01	0.13	1.778	3.56	25	impervious	100
C18	RG1	133731	0.88	60.23	128.68	1.97	0.01	0.13	1.778	3.56	25	impervious	100
C19	RG1	133723001	0.49	92.04	60.49	1.12	0.01	0.13	1.778	3.56	25	impervious	100
C20	RG1	133717	0.87	27.58	111.11	0.25	0.01	0.13	1.778	3.56	25	impervious	100
C21	RG1	133713	1.13	77.88	152.81	0.44	0.01	0.13	1.778	3.56	25	impervious	100
C22	RG1	133719	1.29	63.18	70.716	0.35	0.01	0.13	1.778	3.56	25	impervious	100
C23	RG1	133763	0.74	98	14.89	0.61	0.01	0.13	1.778	3.56	25	impervious	100
C24	RG1	133707	0.35	98	13.172	0.37	0.01	0.13	1.778	3.56	25	impervious	100
C25	RG1	133733	0.34	98	15.37	0.74	0.01	0.13	1.778	3.56	25	impervious	100

Appendix 2. Raw data for the junctions and conduits of the sewer network.

Junction	Elevation (CD		Max depth
Outfall	35.69		
133701	39.82	41.85	2.03
133703	39.62	41.74	2.12
133705	39.33	41.51	2.18
133707	37.8	41.28	3.48
133729	37.68	41.69	4.01
133731	37.58	40.91	3.33
133733	37.51	40.15	2.64
133735	37.45	40.01	2.56
133751	37	39.62	2.62
133753	36.47	39.13	2.66
133755	36.31	38.91	2.6
133757	36.19	38.69	2.5
133759	36.08	38.44	2.36
133761	35.96	38.29	2.33
133763	35.7	37.98	2.28
133723	38.56	40.04	1.48
1.34E+08	38.43	40.27	1.84
133721	38.43	40.44	2.01
133719	38.33	40.58	2.25
133717	38.24	40.89	2.65
133715	38.22	41.03	2.81
133713	38.17	41.09	2.92
133711	37.97	40.95	2.98
133709	37.86	41.09	3.23
133749	40.93	43.35	2.42
133747	39.3	41.65	2.35
133745	39.01	41.28	2.27
133743	39.09	41.18	2.09
133741	38.82	40.98	2.16
133739	38.51	40.49	1.98

Conduits	From node	To node	Length	Roughness	Inoffset	Outoffset	Shape	Geom1
133701	133701	133703	17.72	0.012			Circular	0.3
133703	133703	133705	46.24	0.012			Circular	0.4
133705	133705	133707	60.94	0.012			Circular	0.4
133723	133723	1.34E+08	7.79	0.012			Circular	0.5
1.34E+08	1.34E+08	133721	43.18	0.012			Circular	0.5
133721	133721	133719	24.63	0.012			Circular	0.5
133719	133719	133717	60.86	0.012			Circular	0.6
133717	133717	133715	12.91	0.012			Circular	0.6
133715	133715	133713	15.48	0.012			Circular	0.7
133713	133713	133711	53.83	0.012			Circular	0.7
133711	133711	133709	41.78	0.012			Circular	0.8
133709	133709	133707	44.63	0.012			Circular	0.8
133707	133707	133729	44.47	0.012	0.97		Circular	0.9
133729	133729	133731	49.15	0.012			Circular	0.9
133731	133731	133733	30.04	0.012			Circular	0.9
133733	133733	133735	13.23	0.012			Circular	0.9
133749	133749	133747	29.78	0.012			Circular	0.3
133747	133747	133745	44.94	0.012	0.04		Circular	0.4
133745	133745	133741	46.44	0.012			Circular	0.4
133743	133743	133741	63.98	0.012			Circular	0.4
133741	133741	133739	63.09	0.012			Circular	0.6
133739	133739	133737	58.34	0.012			Circular	0.6
133737	133737	133735	27.92	0.012			Circular	0.6
133735	133735	133751	68.24	0.012	0.68		Circular	0.9
133751	133751	133753	70.66	0.012			Circular	0.9
133753	133753	133755	26.37	0.012	1.05	0.02	Circular	1
133755	133755	133757	26.52	0.012			Circular	1
133757	133757	133759	25.52	0.012			Circular	1
133759	133759	133761	15.18	0.012			Circular	1
133761	133761	133763	59.09	0.012	0.6		Circular	1.2
133763	133763	R33765	20.04	0.012			Circular	1.2

Appendix 3. Raw data for the Rainfall and flow for both the events

Name	Date	Time	mm/20m	Time	Qzu [L/s]
PN2	07-05-2023	06:15	0	6:45	0.00
PN2	07-05-2023	06:35	0.3	6:50	6.98
PN2	07-05-2023	06:55	1.5	7:00	12.04
PN2	07-05-2023	07:15	1.2	7:10	46.98
PN2	07-05-2023	07:35	2.3	7:20	53.43
PN2	07-05-2023	07:55	2.3	7:30	55.26
PN2	07-05-2023	08:15	0.8	7:40	78.77
PN2	07-05-2023	08:35	0.2	7:50	100.05
PN2	07-05-2023	08:55	1.3	8:00	156.76
PN2	07-05-2023	09:15	1.8	8:10	125.65
PN2	07-05-2023	09:35	1.3	8:20	113.81
PN2	07-05-2023	09:55	0	8:30	56.34
PN2	07-05-2023	10:15	0	8:40	29.13
PN2	07-05-2023	10:35	0	8:50	19.35
PN2	07-05-2023	10:55	0	9:00	34.64
PN2	07-05-2023	11:15	0	9:10	37.16
PN2	07-05-2023	11:35	0	9:30	72.67
PN2	07-05-2023	11:55	0	9:50	76.29
				10:10	15.76

Name	Date	Time	mm/20m	Time	Qzu [L/s]
PN1	06/22/202	22:50	0	23:20	0
PN1	06/22/202	23:10	0.8	23:25	9.987259
PN1	06/22/202	23:30	1	23:35	17.40711
PN1	06/22/202	23:50	2	23:45	28.21571
PN1	06/23/202	00:10	1	23:55	64.17437
PN1	06/23/202	00:30	0.5	00:05	82.99425
PN1	06/23/202	00:50	0	00:15	49.03391
PN1	06/23/202	01:10	0.5	00:25	42.8902
PN1	06/23/202	01:30	2.8	00:35	28.97431
PN1	06/23/202	01:50	1.8	00:45	22.75045
PN1	06/23/202	02:10	1.8	00:55	13.15236
PN1	06/23/202	02:30	1.8	01:05	8.760919
PN1	06/23/202	02:50	2.3	01:15	4.308523
PN1	06/23/202	03:10	1.2	01:25	46.15359
PN1	06/23/202	03:30	1	01:35	105.3699
PN1	06/23/202	03:50	1.1	01:45	121.4274
PN1	06/23/202	04:10	0.5	01:55	79.36304
PN1	06/23/202	04:30	0.2	02:05	92.20174
PN1	06/23/202	04:50	0	02:15	91.66927
PN1	06/23/202	05:10	0	02:25	91.97633
PN1	06/23/202	05:30	0	02:35	76.61074
PN1	06/23/202	05:50	0	02:45	75.51674
PN1	06/23/202	06:10	0	02:55	75.51674
PN1	06/23/202	06:30	0	03:05	75.51674
PN1	06/23/202	06:50	0	03:15	75.51674
PN1	06/23/202	07:10	0	03:25	75.51674
				03:35	75.51674
				03:45	75.51674
				03:55	75.51674
				04:05	67.83812
				04:15	39.39904
				04:25	28.30072
				04:45	19.58961
				05:05	12.31347
				05:25	7.606325
				05:45	3.662304
				06:05	2.817157